

4MOST

SPV S6 New Approach

Vivek

Group 1:

- fields from eFEDS area observable during Jan - July, with 1600 targets selected as following
 - 1.Subset-1: **40%** selected from SDSS-DESI with CLASS == QSO ($r < 23.8$; 50% with $1 < z < 2$)
 - 2.Subset-2: **30%** selected from eFEDs and SDSS-DESI with CLASS != QSO ($r < 23.8$; 50% with $1 < z < 2$; reject STARS)
 - 3.Subset-3: **30%** with no spectroscopy in SDSS-DESI (use DECALS photometry)
 - 1.50% from eROSITA DR1 (Merloni et al. Messenger article)
 - 2.50% with $r < 23$ and $r - W2 > 5.9$ (Andonie et al.)

Group 2:

Fields outside eFEDS area observable during Jan - July

1.Subset-1: **40%** (out of 1600) selected from SDSS-DESI with CLASS == QSO ($r < 23.8$; 50%; $1 < z < 2$)

2.Sunset-2: **30%** selected from SDSS-DESI with CLASS != QSO ($r < 23.8$; 50%; $1 < z < 2$) but

~~1.Satisfy eROSITA criteria mentioned above.~~

~~2.Satisfy IR criteria mentioned above.~~

3.Subset-3: **30%** with **no spectroscopy** in SDSS-DESI (use DECALS photometry)

1.50% from eROSITA

2.50% with $r < 23$ and $r - W2 > 5.9$ (use unWISE)

Removed
This

Prioritise this whenever possible

eFEDS extragalactic catalog as per Andrea instructions

Set 1 40%

Master catalog of SDSS
and DESI CLASS =
QSO

With spectra

Set 2 30%

Master catalog of SDSS
and DESI CLASS !
=QSO + ~~Set3 overlap~~

With spectra

High z
catalog
Yang

Set 3

eRASS1 catalog
X-ray selection

30%

Andone et al
catalog WISE
selection

No spectra

R mag < 23.8

50% $1 < z < 2$

Main Filters are : ZWARNING = 0, DEC < 10 and RA > 30

Common
with eFEDs

Summary Table : Inside eFEDS

ID	RA_center	DEC_center	sel1 total	sel1 common	sel1 noncommon	sel2 total	sel3 total	sel3 eRASS	sel3 unwise	grand total
eFEDS_1	144.001535	3.027709	640	189	451	480	480	136	344	1600
eFEDS_2	141.501535	3.027709	640	181	459	480	480	134	346	1600
eFEDS_3	132.001535	1.427709	640	311	329	480	480	137	343	1600
eFEDS_4	139.201535	1.155000	640	390	250	480	480	133	347	1600
eFEDS_5	136.901535	-0.957229	640	137	503	480	480	125	355	1600
eFEDS_6	143.301535	0.609000	640	165	475	480	480	114	366	1600
eFEDS_7	129.530000	4.454000	640	102	538	480	480	133	347	1600
eFEDS_8	132.250000	3.859000	640	105	535	480	480	129	351	1600
eFEDS_9	135.850000	1.111000	640	185	455	480	480	122	358	1600
eFEDS_10	138.630000	3.852000	640	138	502	480	480	131	349	1600

This table gives how many percentage of targets are selected from each Selection. Everything is as required

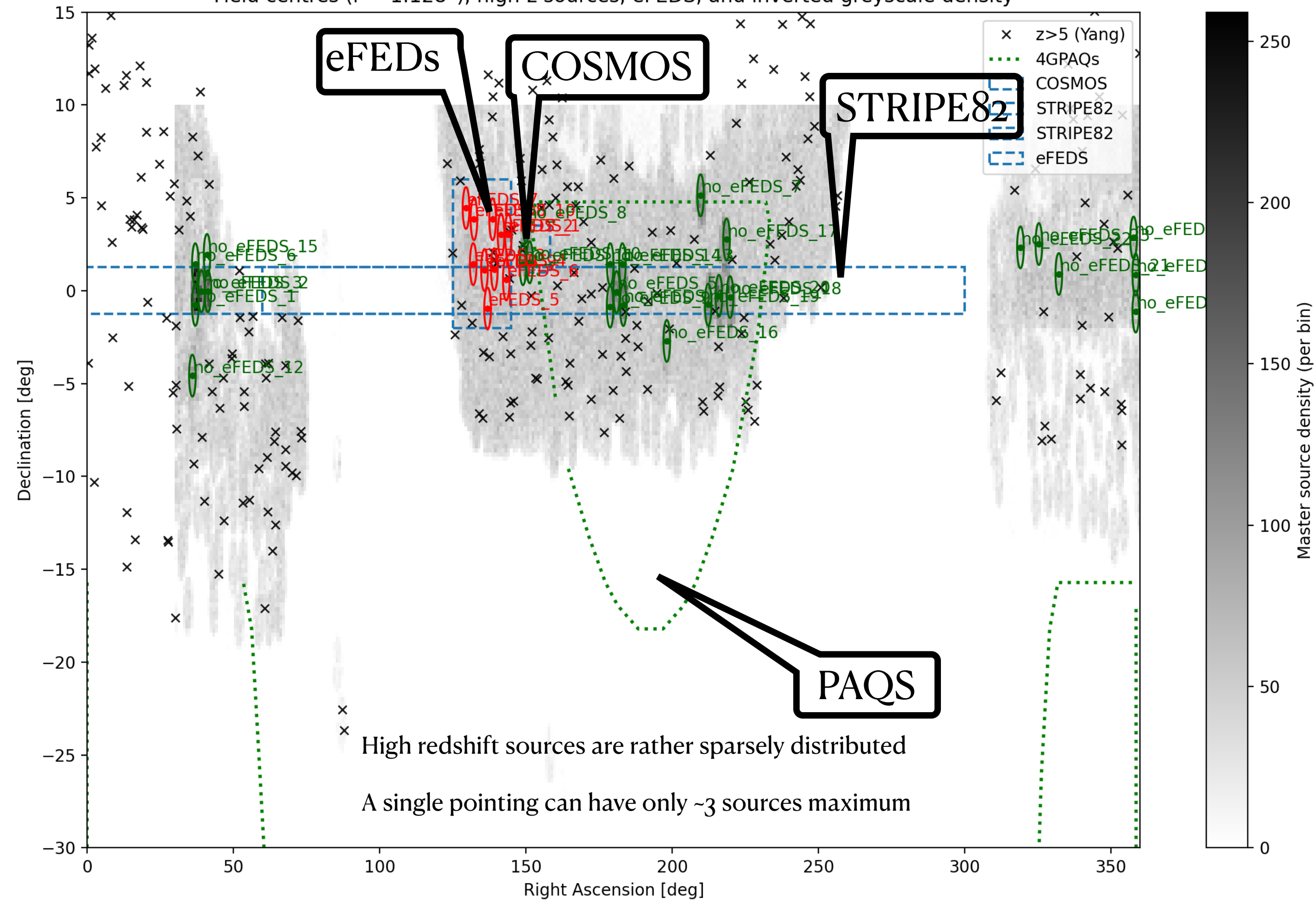
Summary Table : Inside eFEDS

ID	RA_center	DEC_center	set1 count	set1_%	set2 count	set2_%	set3 count	set3_%	total selected	target total
eFEDS_1	144.001535	3.027709	800	40.0	220	30.0	480	30.0	1600	1600
eFEDS_2	141.501535	3.027709	800	40.0	287	30.0	480	30.0	1600	1600
eFEDS_3	132.001535	1.427709	800	40.0	282	30.0	480	30.0	1600	1600
eFEDS_4	139.201535	1.155000	800	40.0	310	30.0	480	30.0	1600	1600
eFEDS_5	136.901535	-0.957229	800	40.0	320	30.0	480	30.0	1600	1600
eFEDS_6	143.301535	0.609000	800	40.0	295	30.0	480	30.0	1600	1600
eFEDS_7	129.530000	4.454000	800	40.0	299	30.0	480	30.0	1600	1600
eFEDS_8	132.250000	3.859000	800	40.0	282	30.0	480	30.0	1600	1600
eFEDS_9	135.850000	1.111000	800	40.0	291	30.0	480	30.0	1600	1600
eFEDS_10	138.630000	3.852000	800	40.0	273	30.0	480	30.0	1600	1600

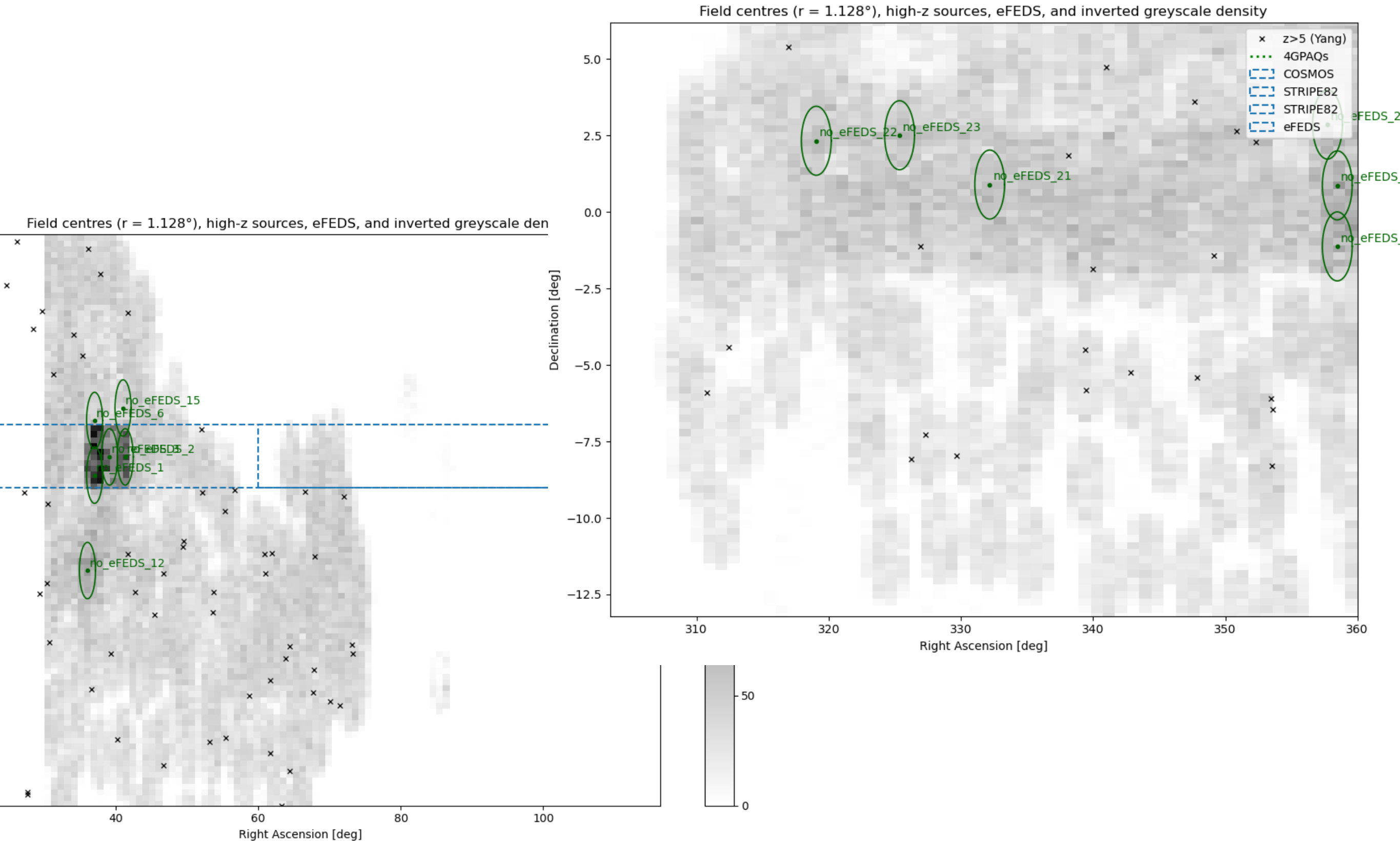
Field	RA_center	DEC_center	sel1 total	sel1 mm	sel1 concommer	sel2 total	sel3 total	sel3 ePASS	sel3 unwise	grand total	
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no_eFEDS_2	41.301535	-0.027229	640	0	640	480	480	0	480	1600	
no_eFEDS_3	39.101535	-0.027229	640	0	640	480	480	0	480	1600	
no_eFEDS_4	183.301535	-0.772291	640	0	640	480	480	198	282	1600	PAQS/S82
no_eFEDS_5	181.001535	-0.072291	640	0	640	480	480	165	315	1600	PAQS/S82
no_eFEDS_6	37.001535	1.427709	640	0	640	480	480	0	480	1600	
no_eFEDS_7	209.701535	5.137709	640	0	640	480	480	186	294	1600	
no_eFEDS_8	150.001535	3.727709	640	0	640	480	480	106	374	1600	COSMOS
no_eFEDS_9	178.801535	-0.857229	640	0	640	480	480	156	324	1600	PAQS
no_eFEDS_10	151.301535	1.527709	640	0	640	480	480	130	350	1600	COSMOS
no_eFEDS_11	149.001535	1.427709	640	0	640	480	480	107	373	1600	COSMOS
no_eFEDS_12	36.001535	-4.572291	640	0	640	480	480	0	480	1600	
no_eFEDS_13	183.001535	1.427709	640	0	640	480	480	182	298	1600	PAQS
no_eFEDS_14	178.831535	1.427709	640	0	640	480	480	161	319	1600	
no_eFEDS_15	41.001535	1.927709	640	0	640	480	480	0	480	1600	
no_eFEDS_16	198.170000	-2.704170	640	0	640	480	480	186	294	1600	PAQS
no_eFEDS_17	218.603243	2.758373	640	0	640	480	480	206	274	1600	PAQS
no_eFEDS_18	219.790775	-0.341460	640	0	640	480	480	238	242	1600	PAQS/S82
no_eFEDS_19	212.314362	-0.749672	640	0	640	480	480	184	296	1600	PAQS/S82
no_eFEDS_20	215.985987	-0.277543	640	0	640	480	480	238	242	1600	PAQS/S82
no_eFEDS_21	332.180000	0.900000	640	0	640	480	480	0	480	1600	
no_eFEDS_22	319.060000	2.332000	640	0	640	480	480	0	480	1600	
no_eFEDS_23	325.360000	2.513000	640	0	640	480	480	0	480	1600	
no_eFEDS_24	358.480000	0.873000	640	0	640	480	480	0	480	1600	
no_eFEDS_25	358.480000	-1.122000	640	0	640	480	480	0	480	1600	
no_eFEDS_26	357.740000	2.861000	640	0	640	480	480	0	480	1600	

Field	RA_center	DEC_center	set1_#	set1_%	set2_#	set2_%	set3_#	set3_%	total_sel	target
no_eFEDS_1	37.001535	-0.757229	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_2	41.301535	-0.027229	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_3	39.101535	-0.027229	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_4	183.301535	-0.772291	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_5	181.001535	-0.072291	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_6	37.001535	1.427709	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_7	209.701535	5.137709	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_8	150.001535	3.727709	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_9	178.801535	-0.857229	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_10	151.301535	1.527709	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_11	149.001535	1.427709	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_12	36.001535	-4.572291	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_13	183.001535	1.427709	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_14	178.831535	1.427709	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_15	41.001535	1.927709	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_16	198.170000	-2.704170	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_17	218.603243	2.758373	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_18	219.790775	-0.341460	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_19	212.314362	-0.749672	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_20	215.985987	-0.277543	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_21	332.180000	0.900000	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_22	319.060000	2.332000	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_23	325.360000	2.513000	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_24	358.480000	0.873000	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_25	358.480000	-1.122000	640	40.0	480	30.0	480	30.0	1600	1600
no_eFEDS_26	357.740000	2.861000	640	40.0	480	30.0	480	30.0	1600	1600

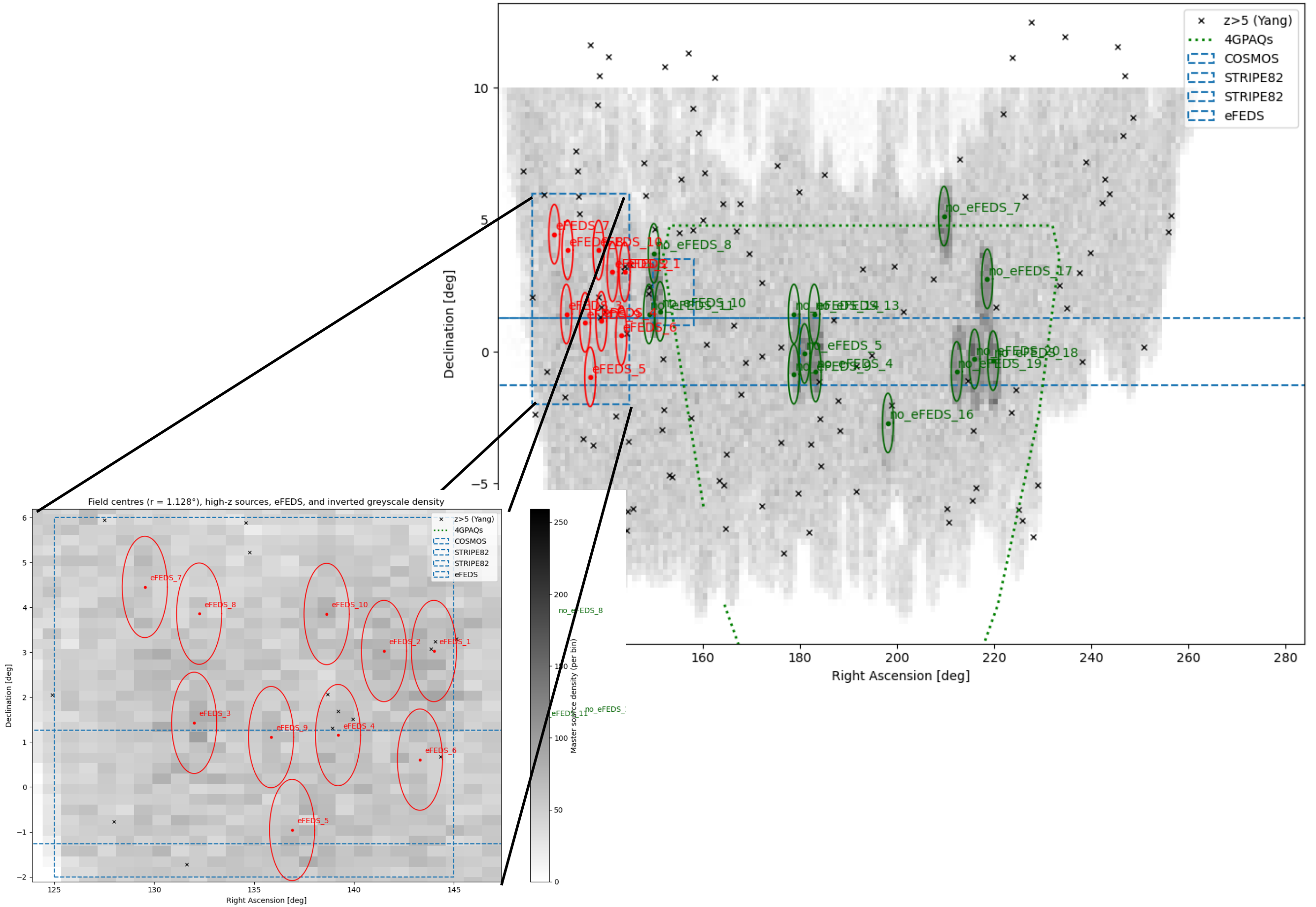
Field centres ($r = 1.128^\circ$), high- z sources, eFEDS, and inverted greyscale density



These 6 fields are to fill out the June July window incase SPV is extended



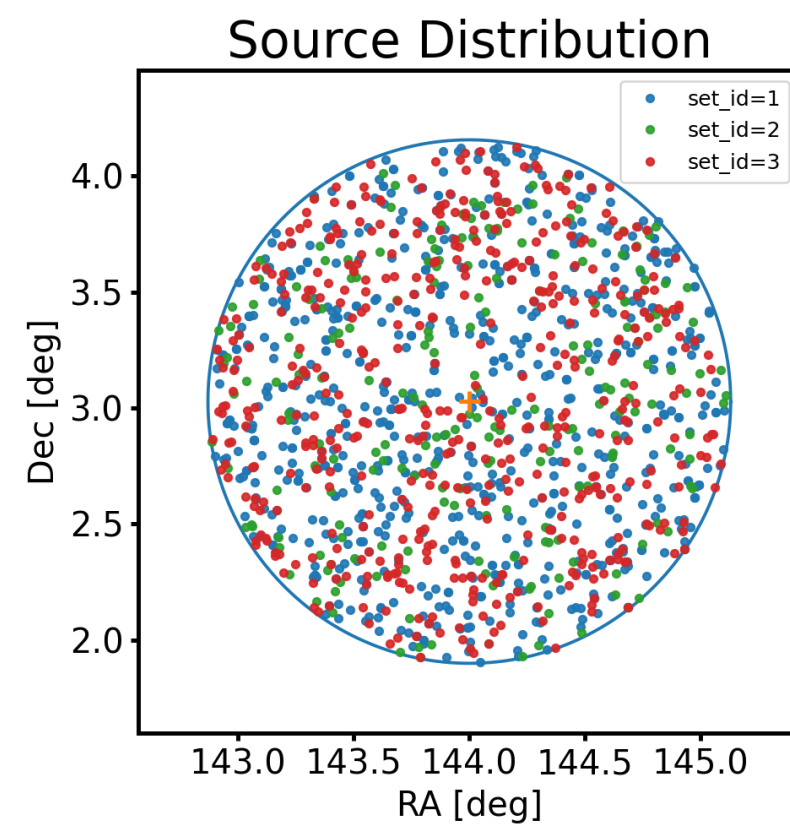
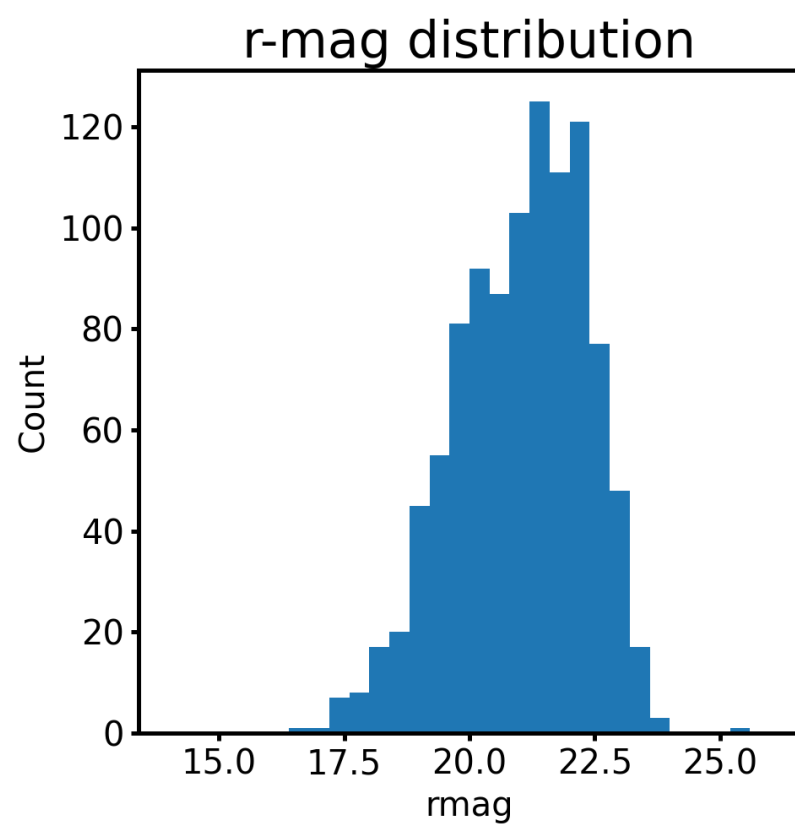
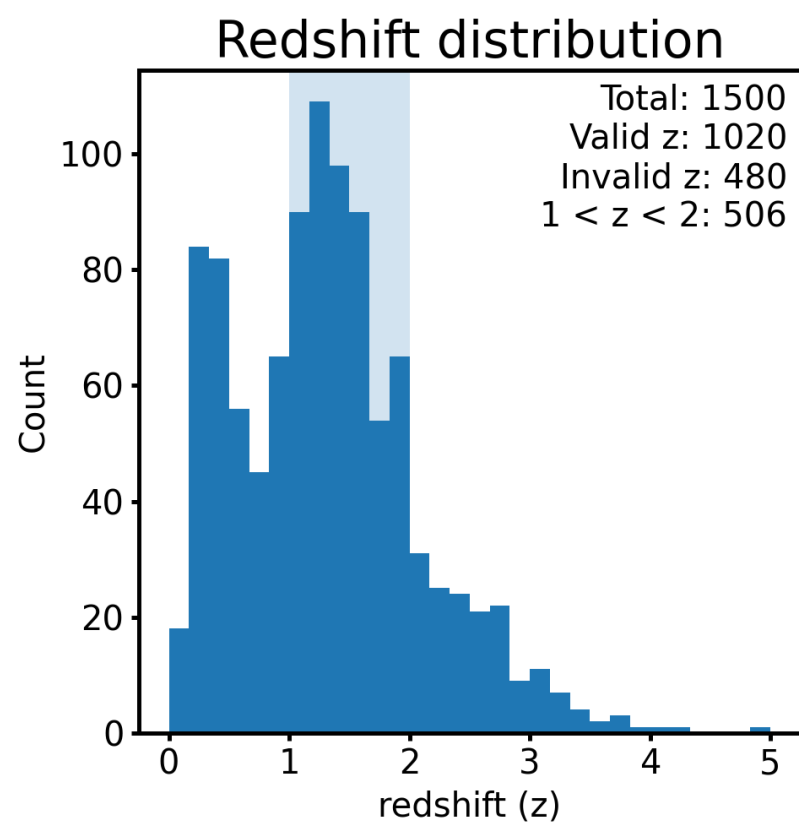
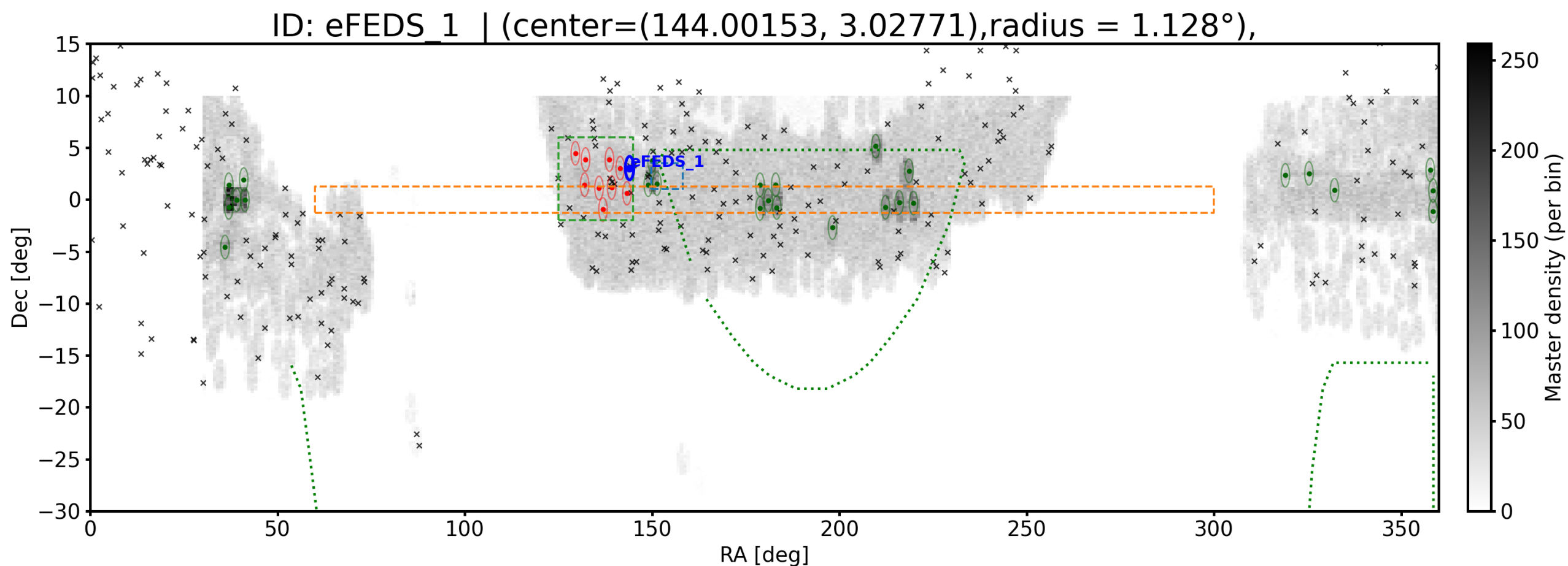
Field centres ($r = 1.128^\circ$), high- z sources, eFEDS, and inverted greyscale density



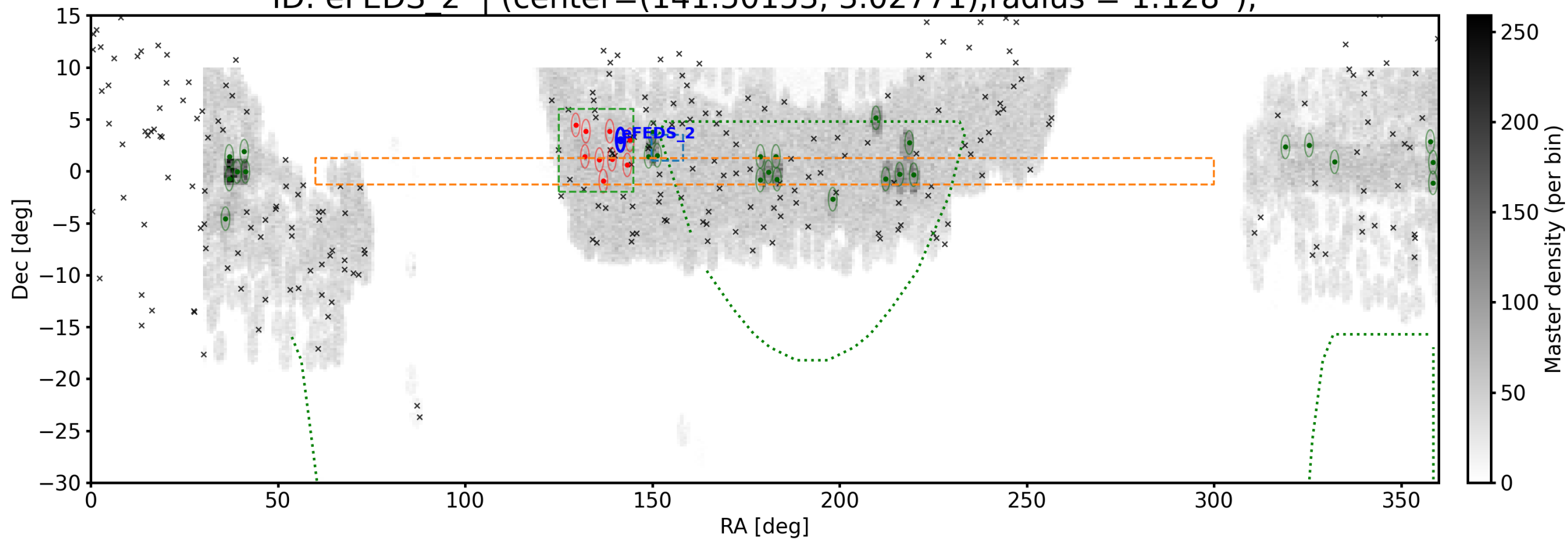
Visibility at Paranal

Paranal, Chile: Year Visibility — Astropy (astronomical night, alt $\geq$ 30°)

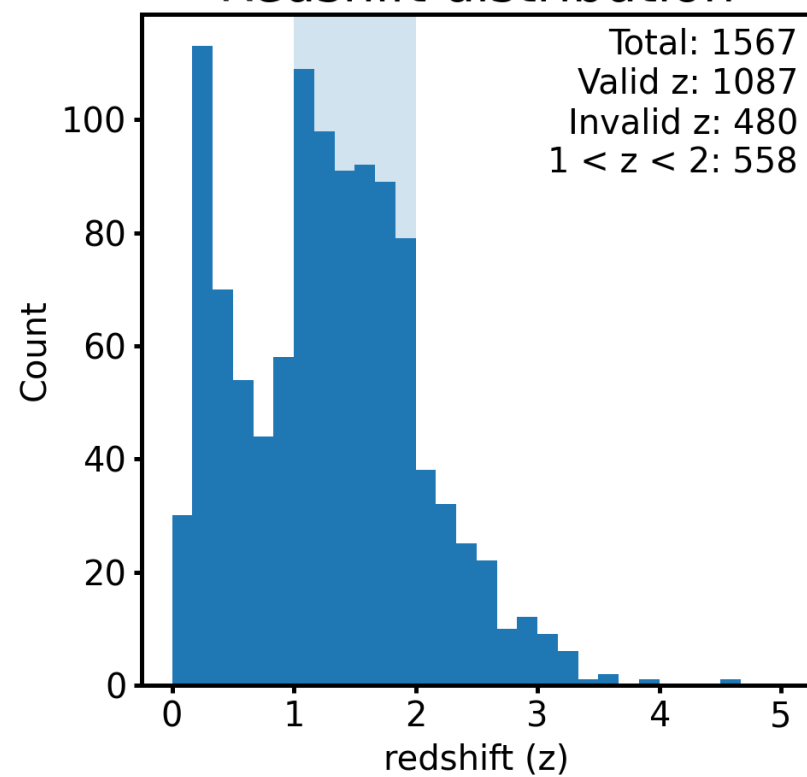




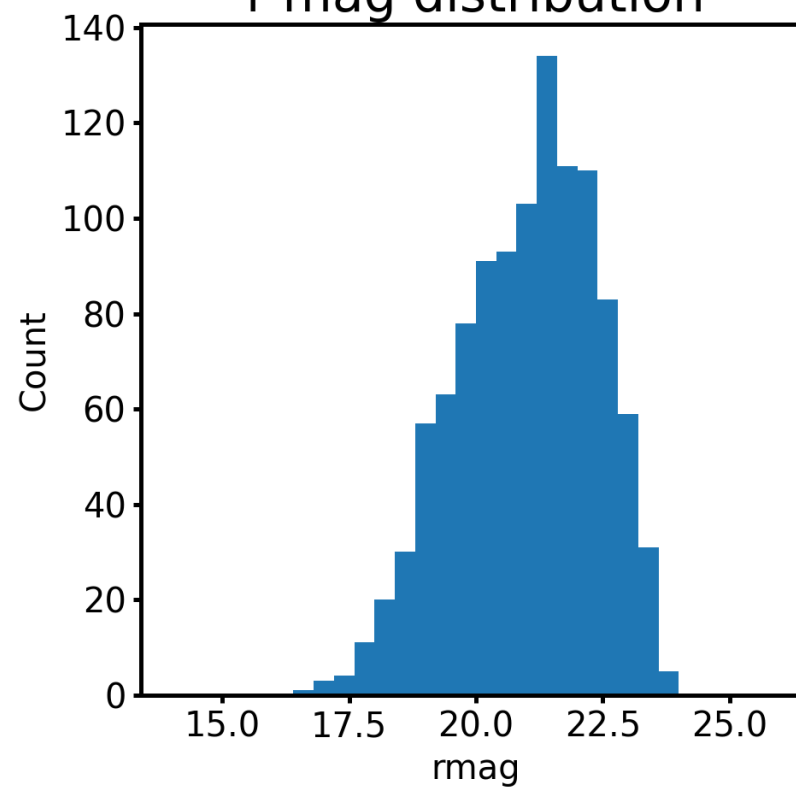
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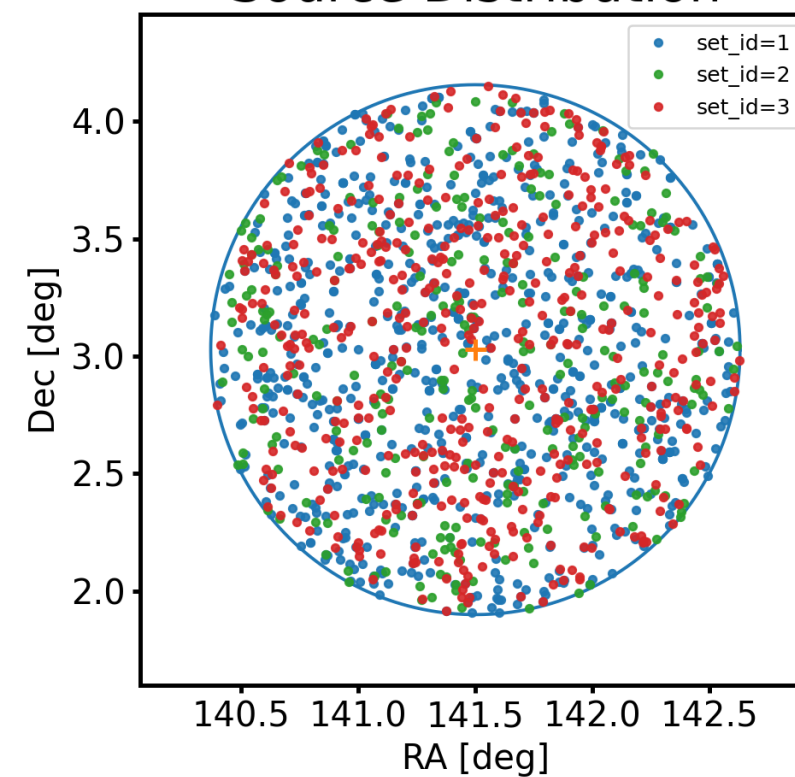
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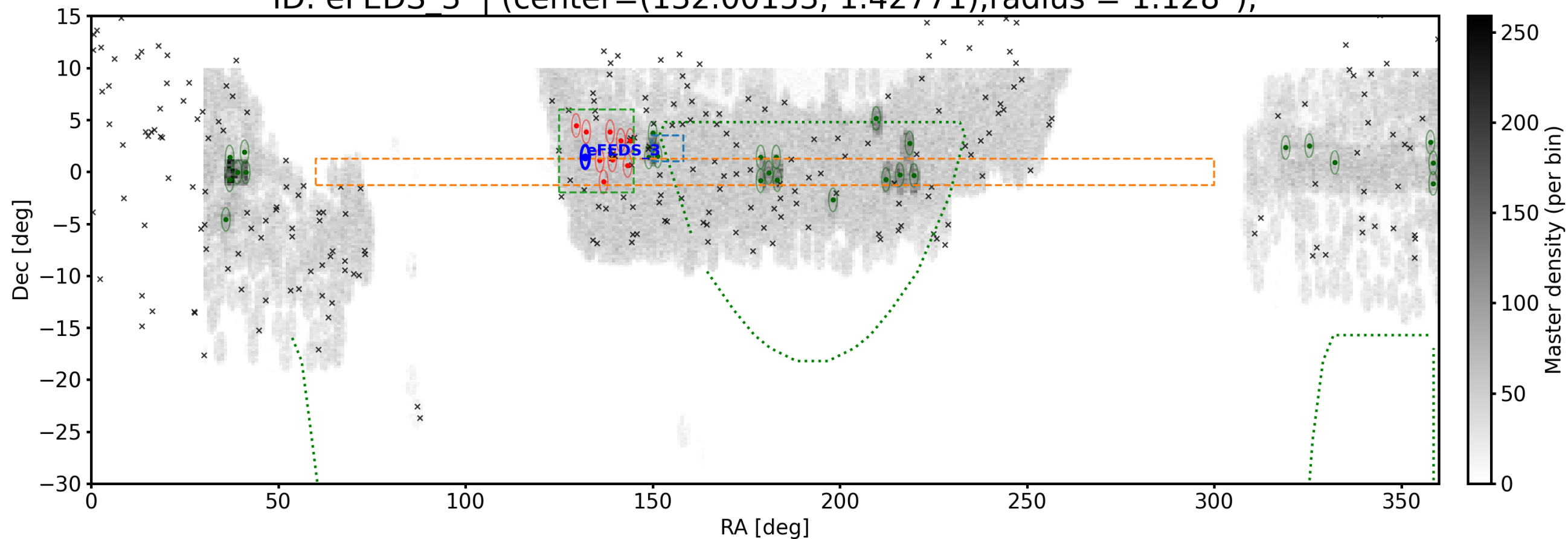
r-mag distribution



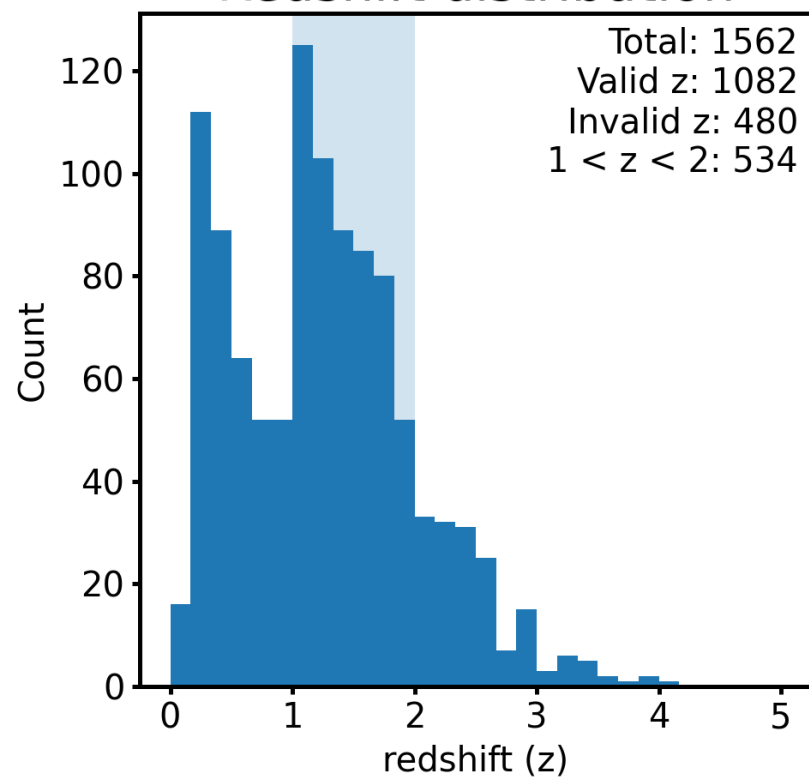
Source Distribution



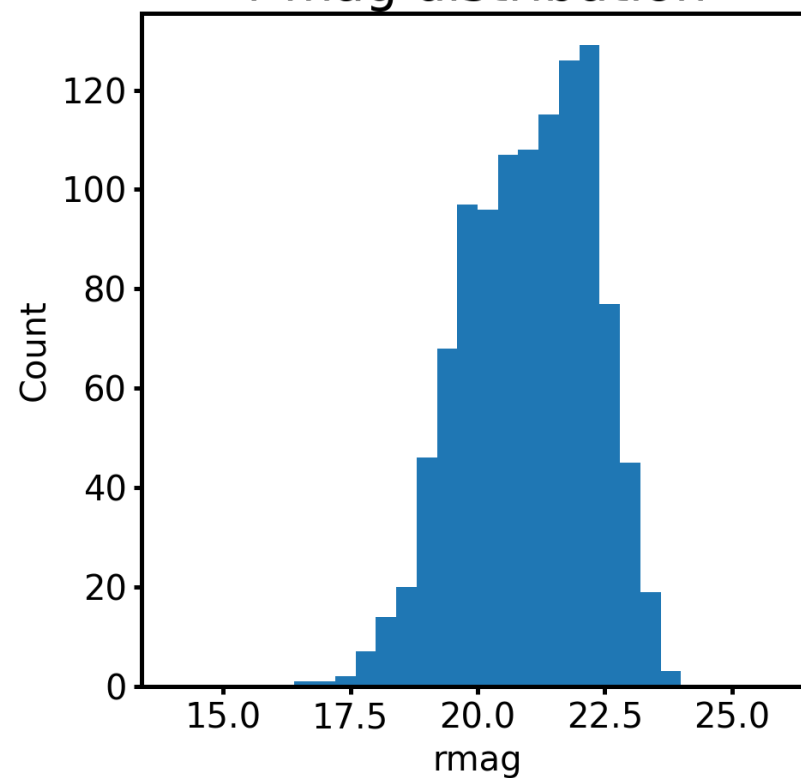
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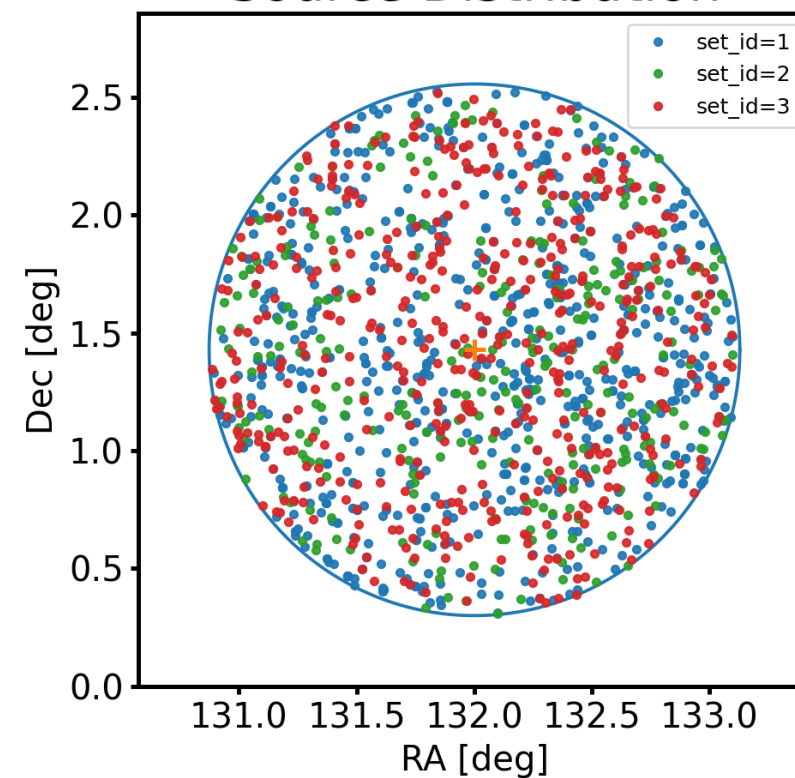
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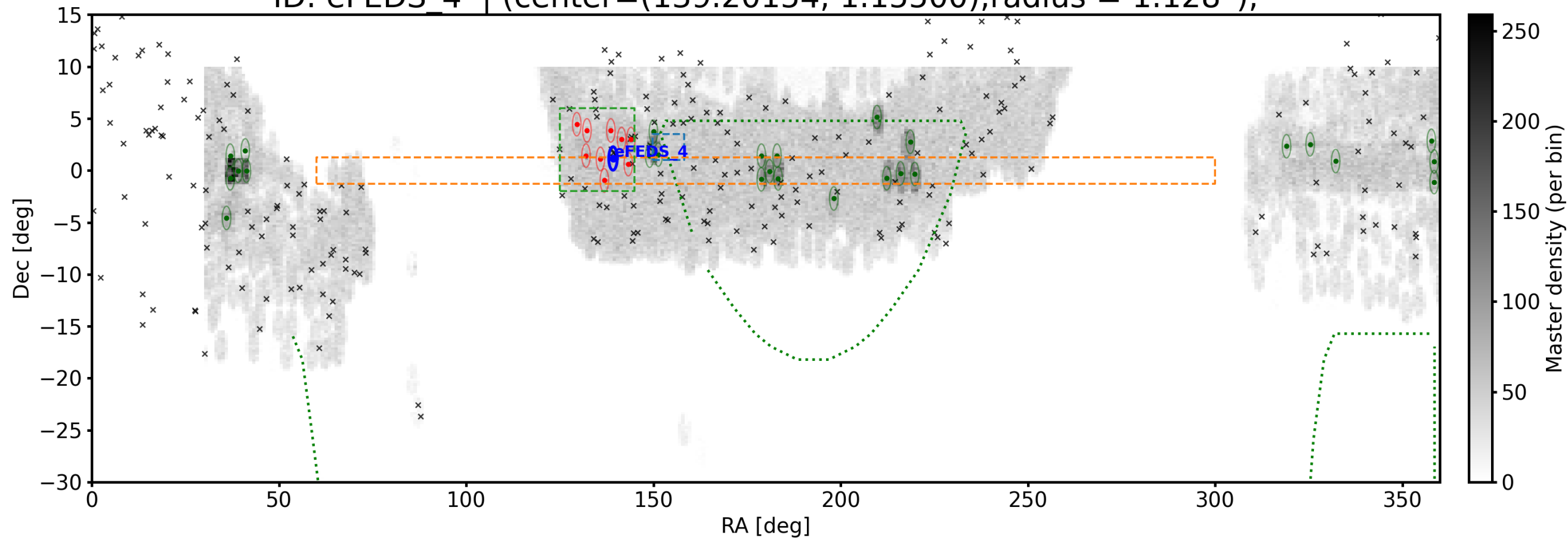
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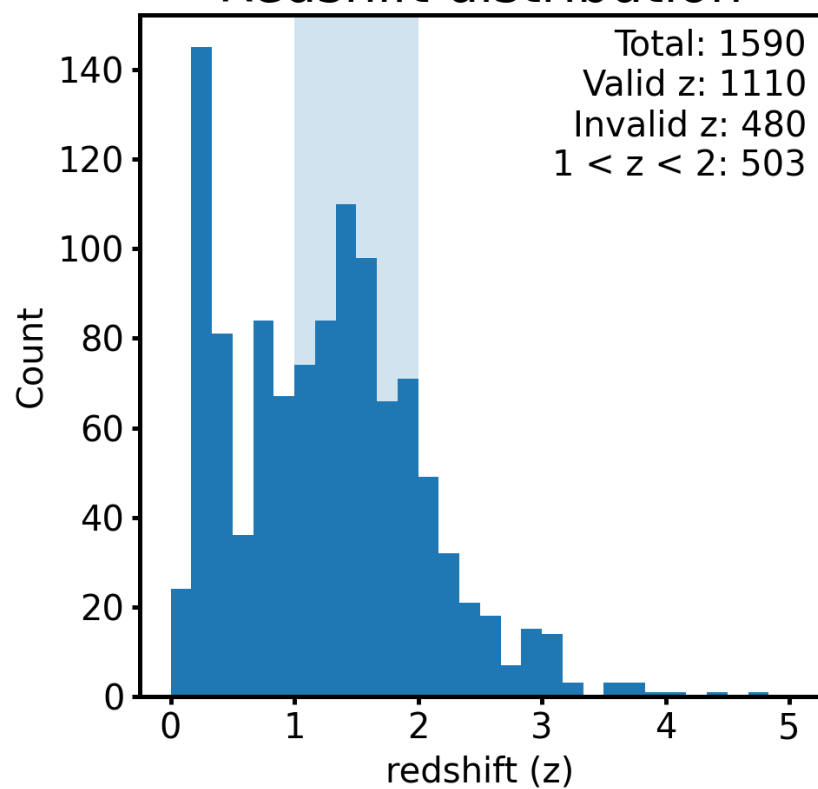
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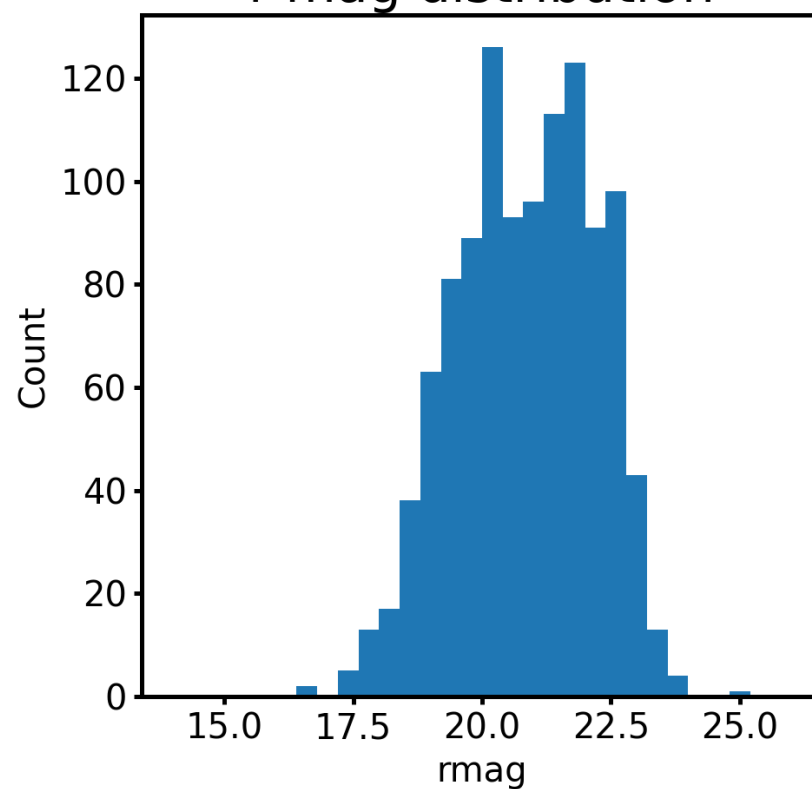
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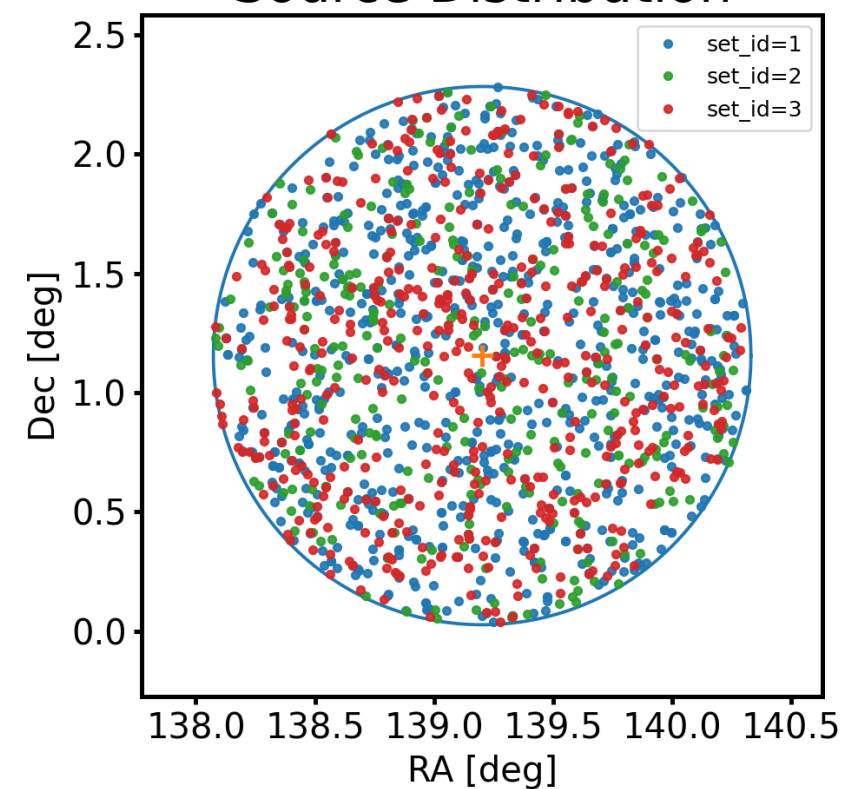
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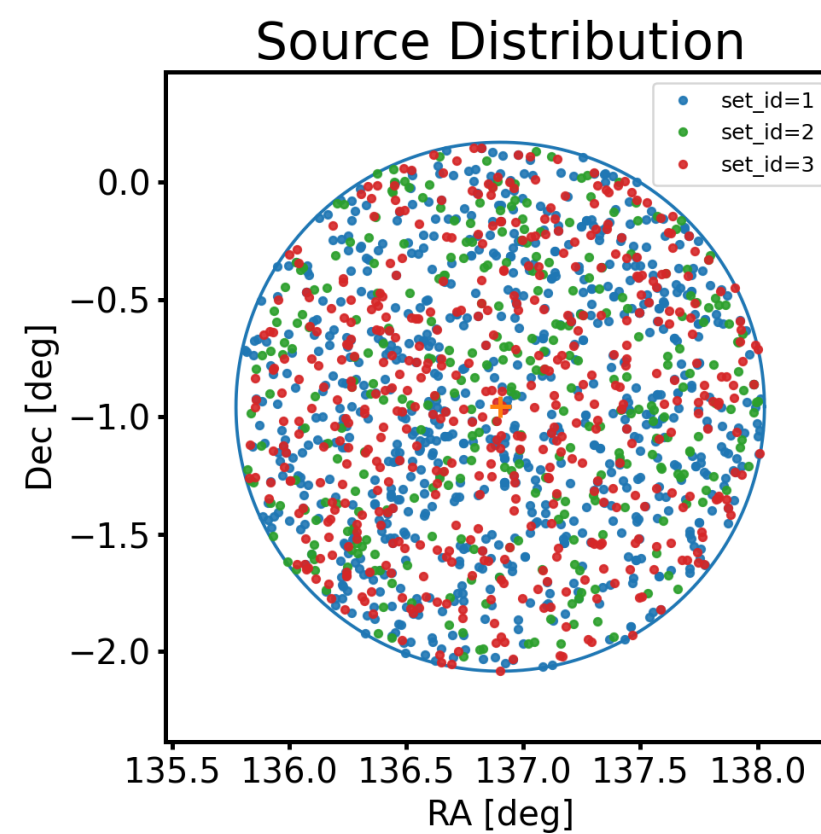
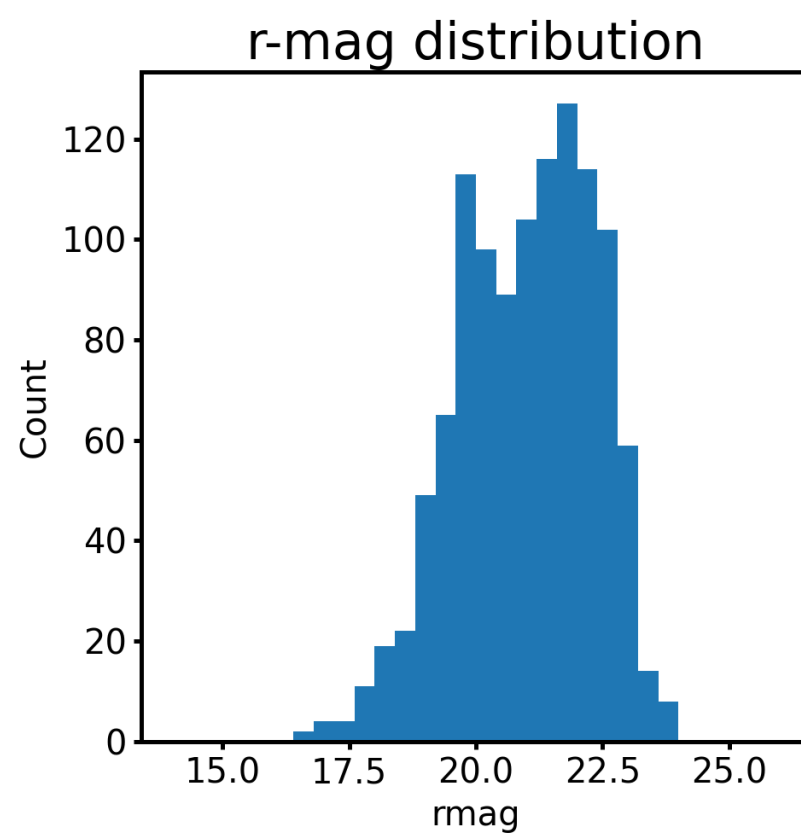
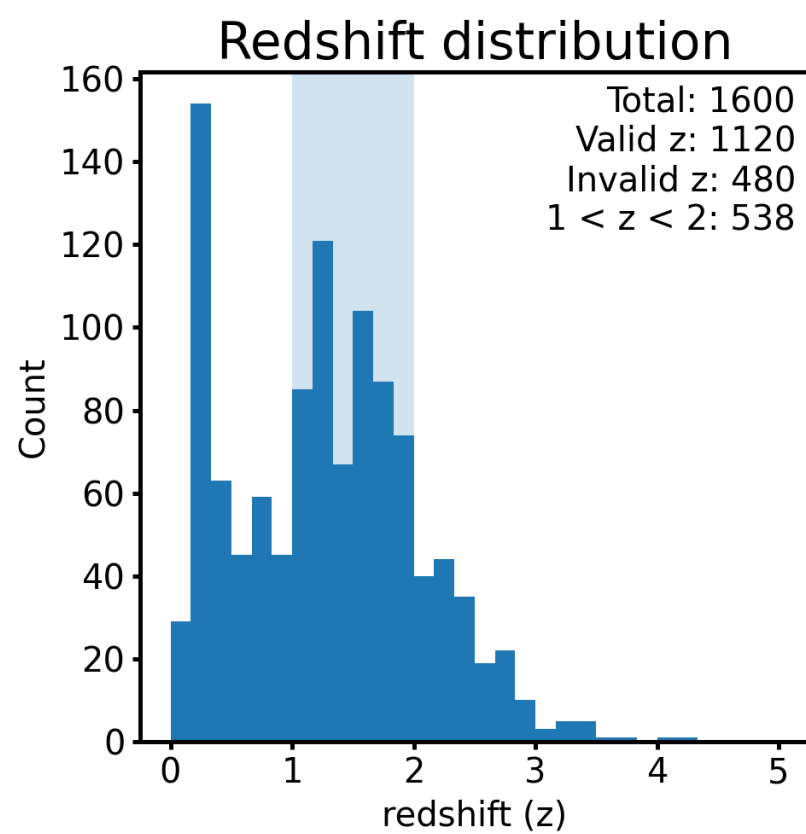
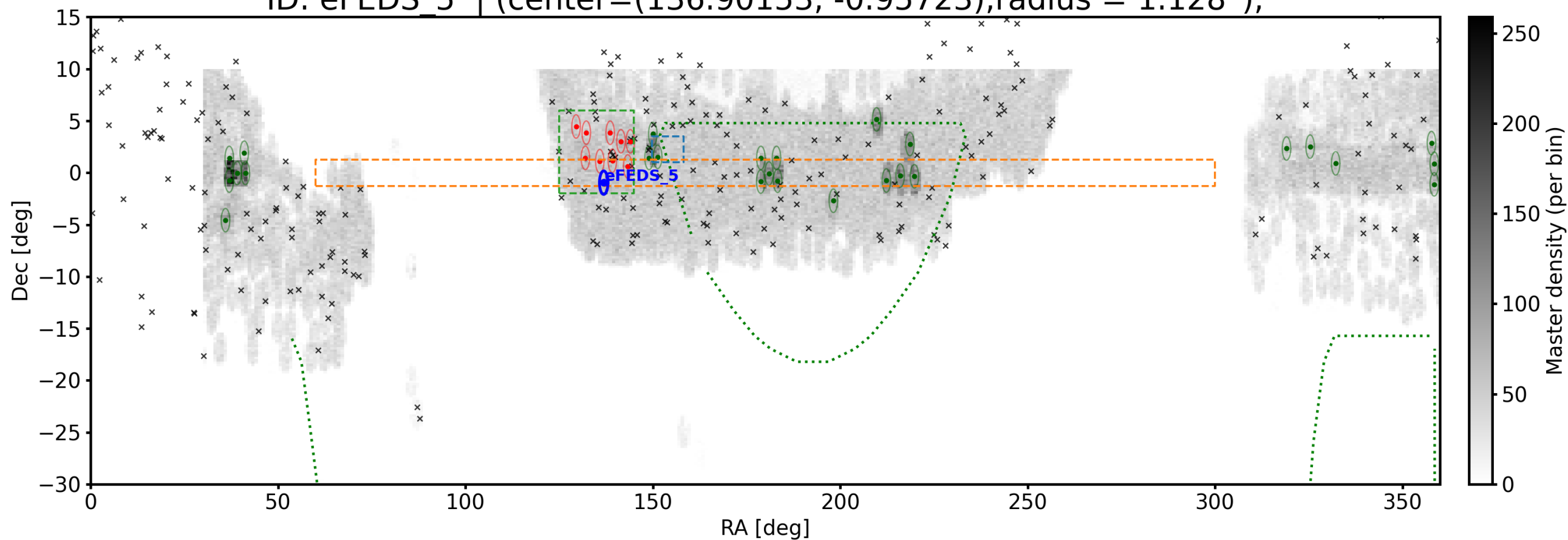
r-mag distribution



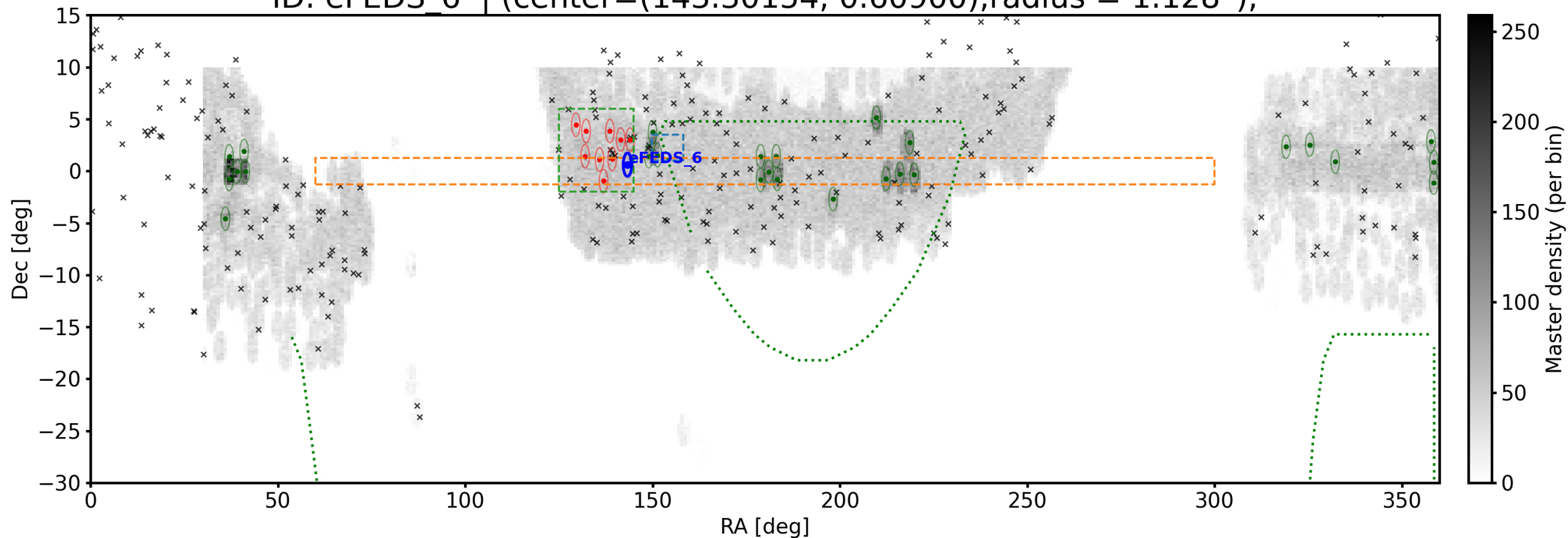
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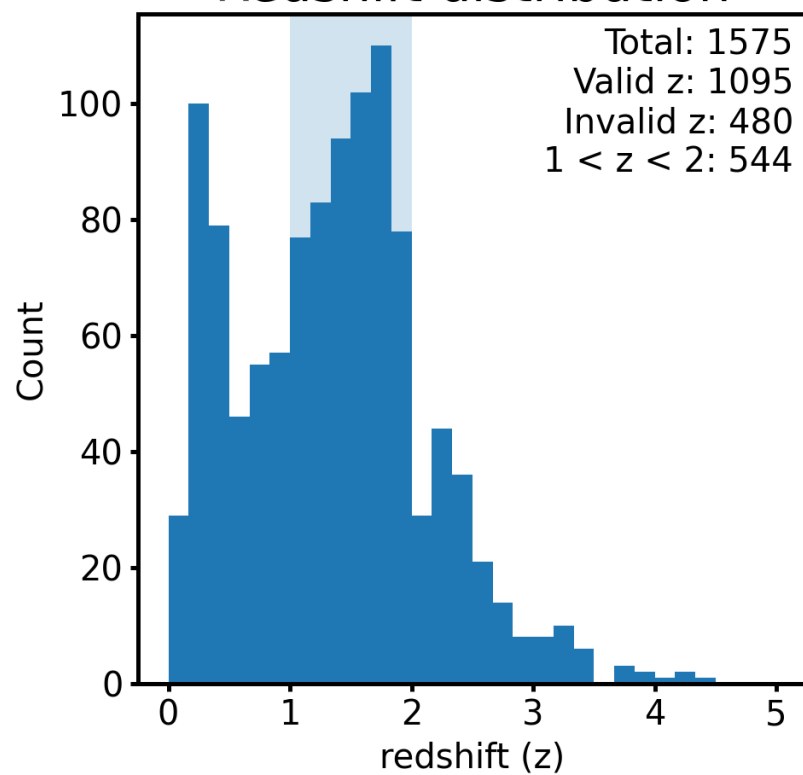
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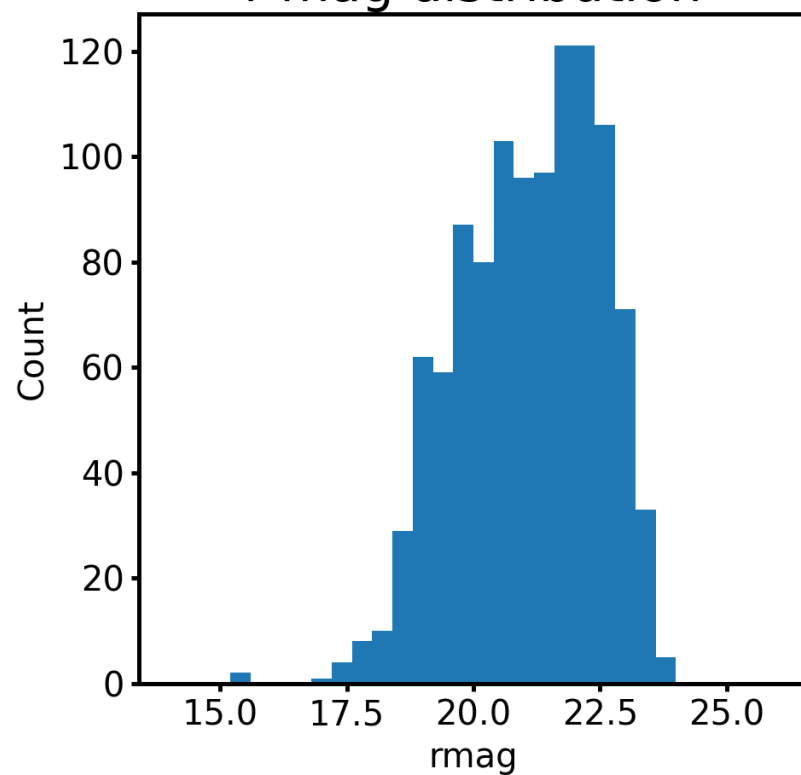
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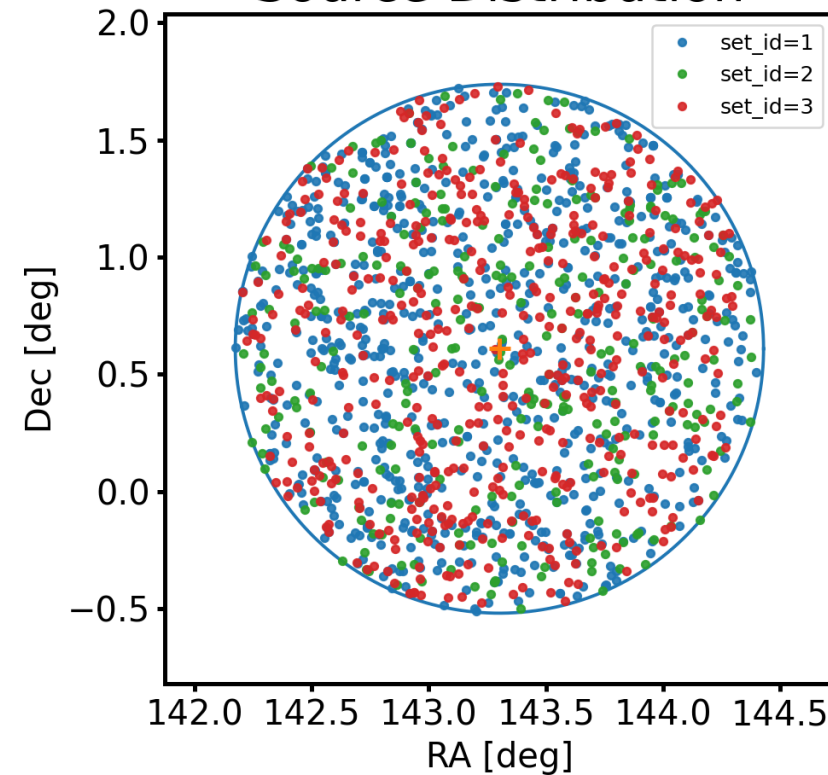
Redshift distribution

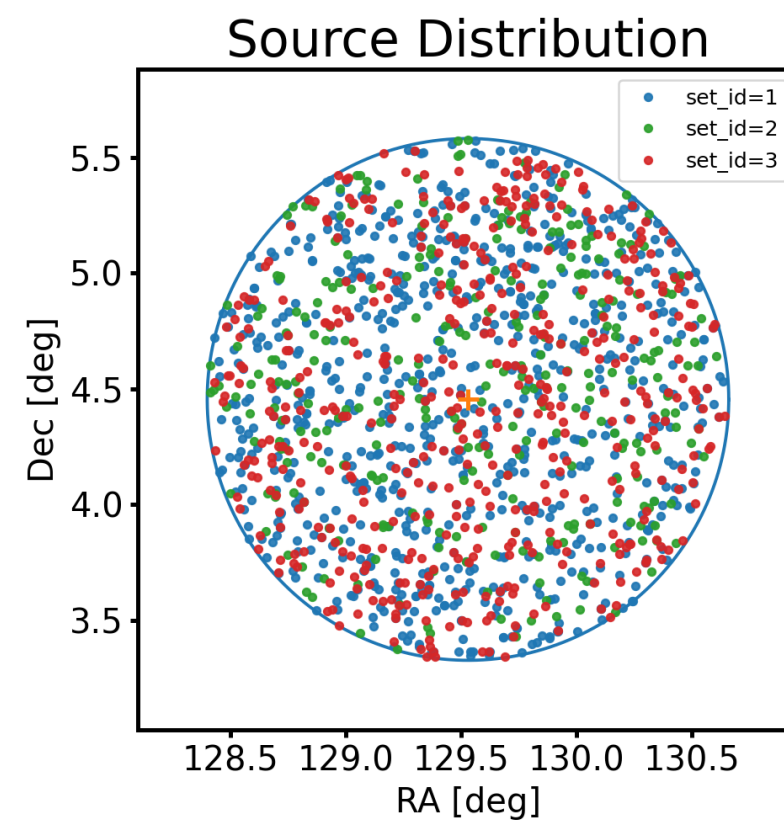
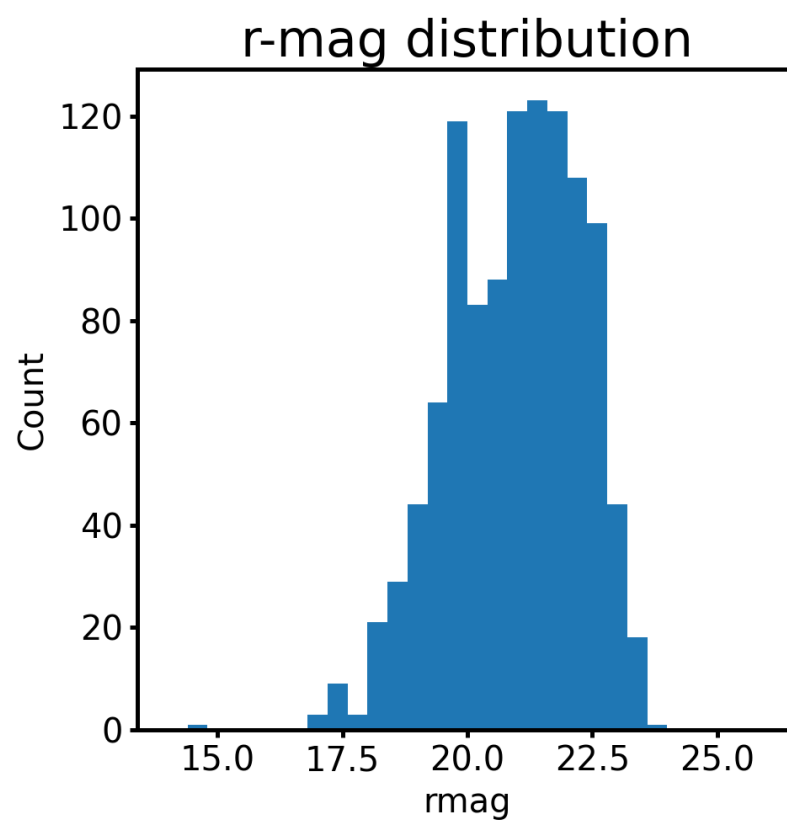
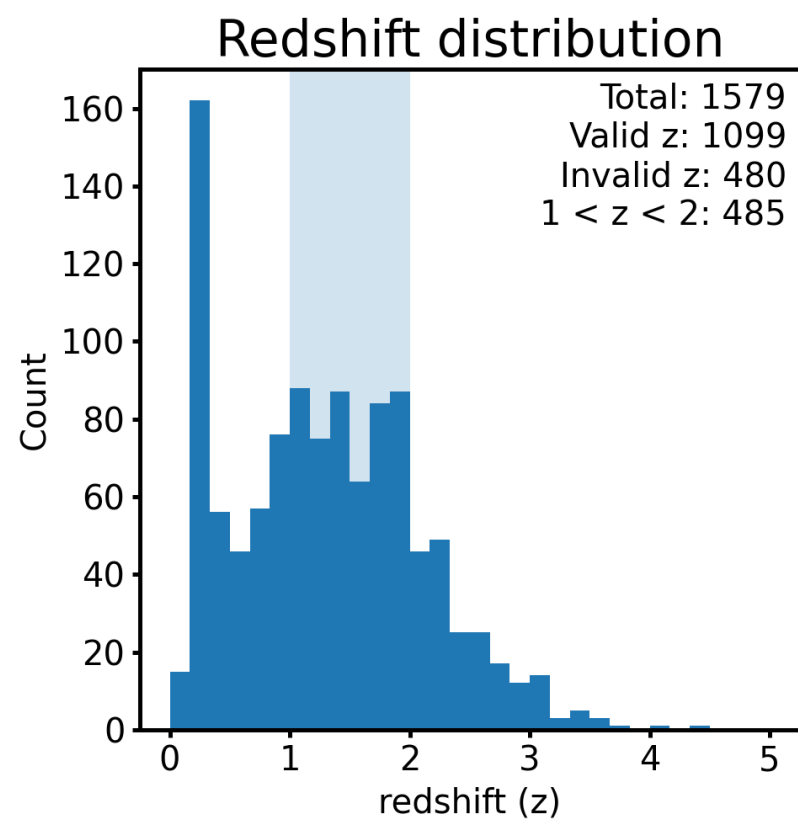
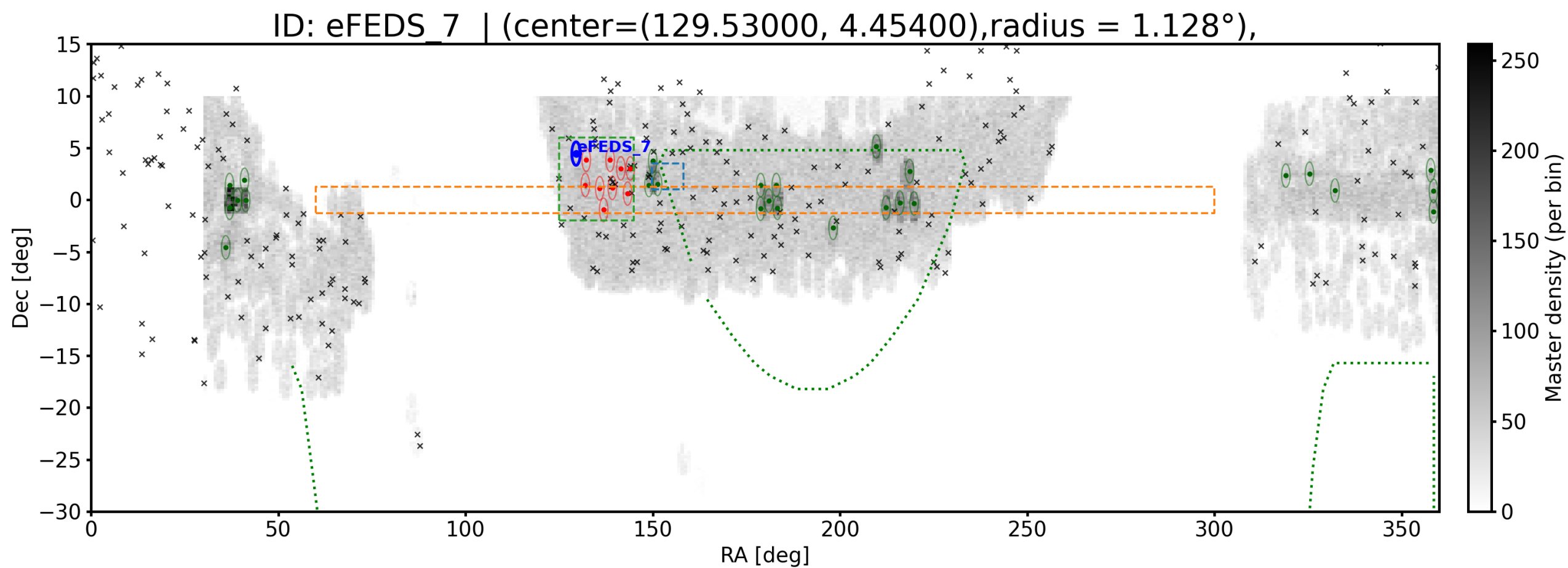


r-mag distribution

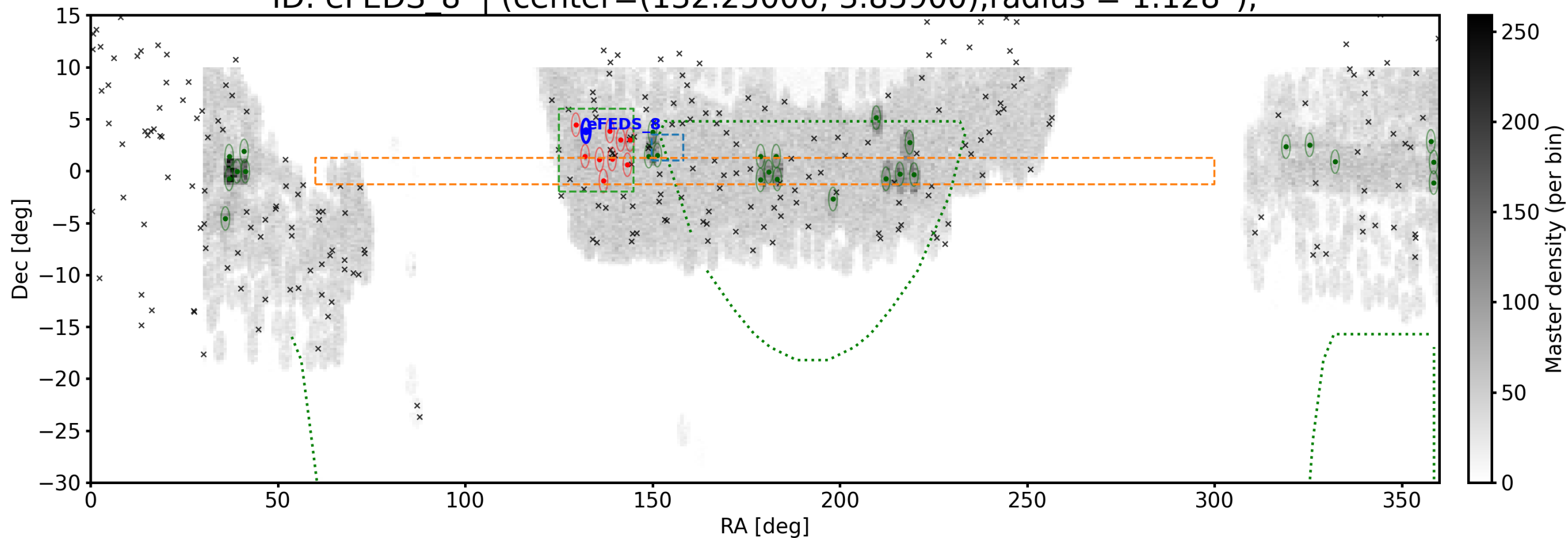


Source Distribution

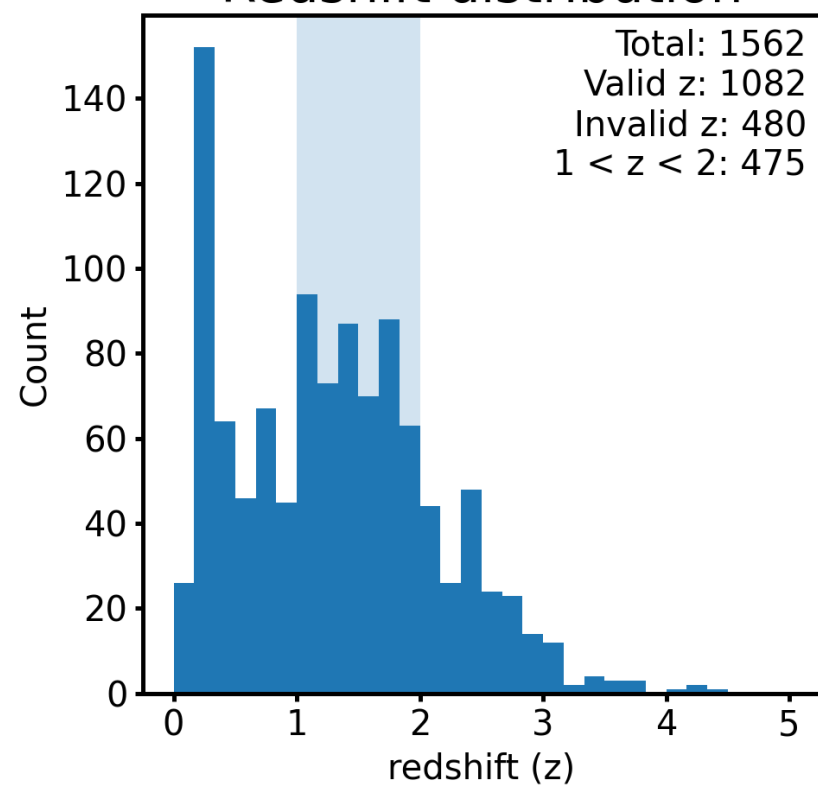




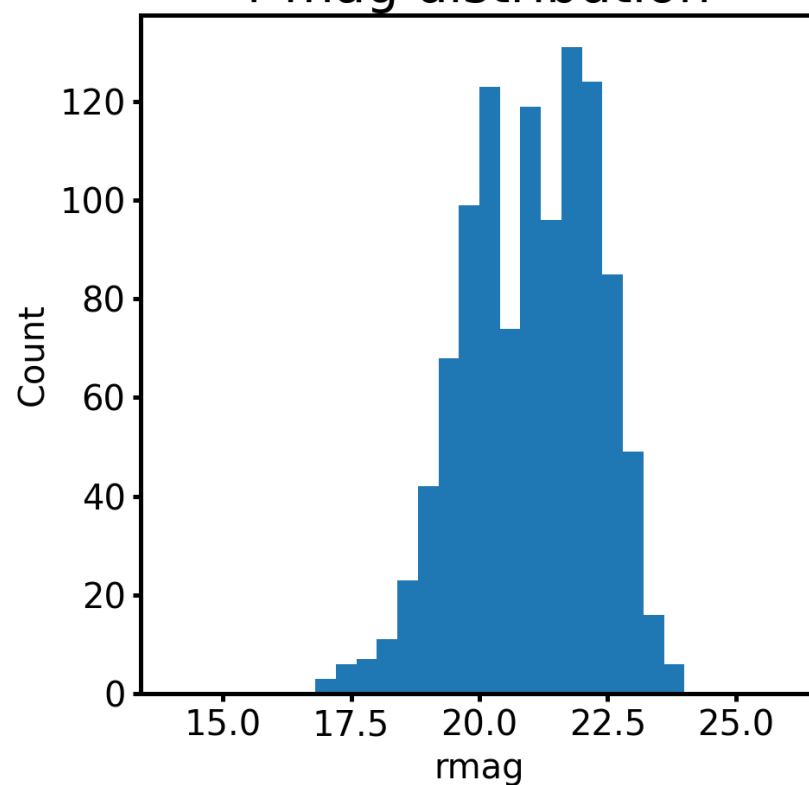
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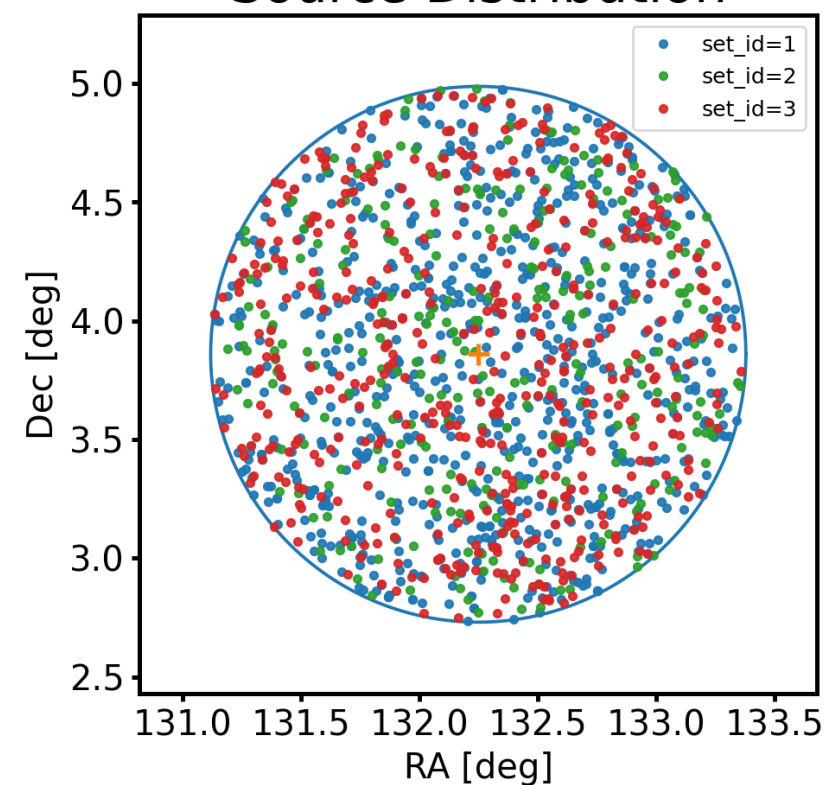
Redshift distribution



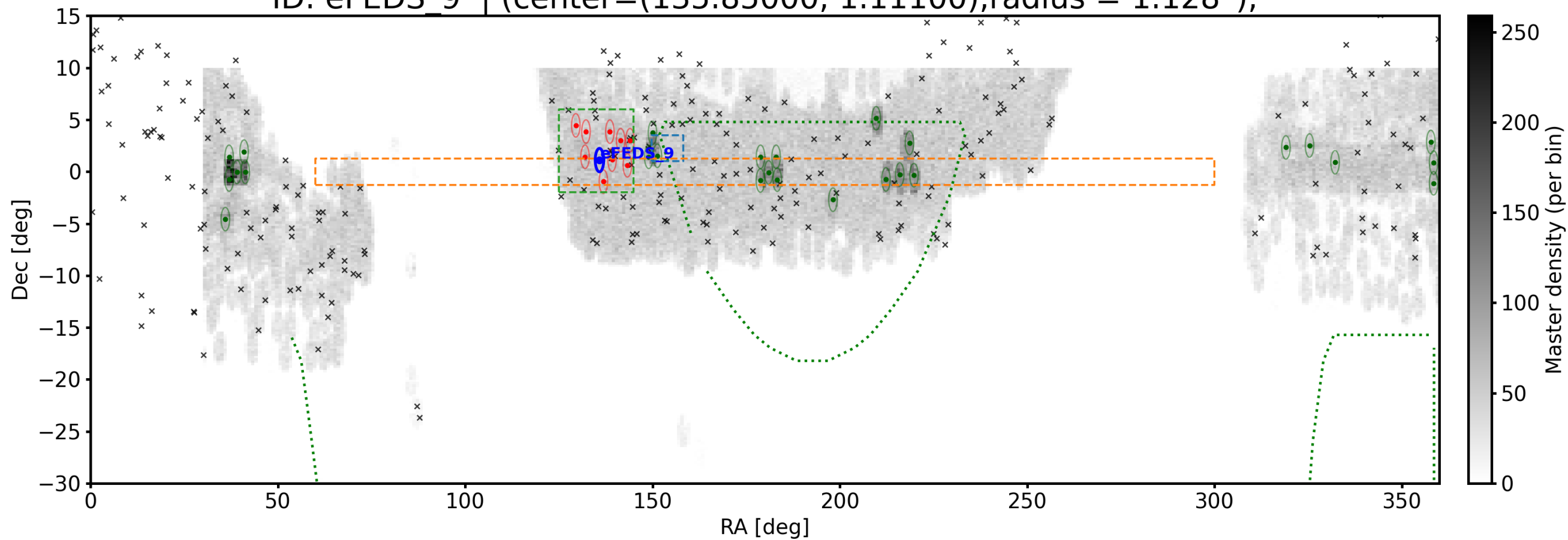
r-mag distribution



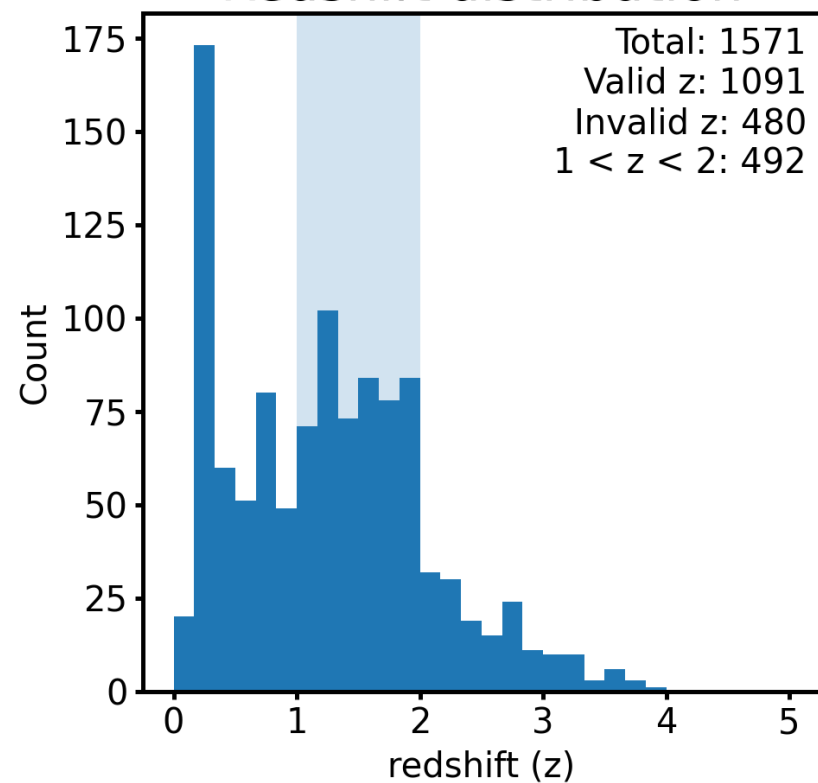
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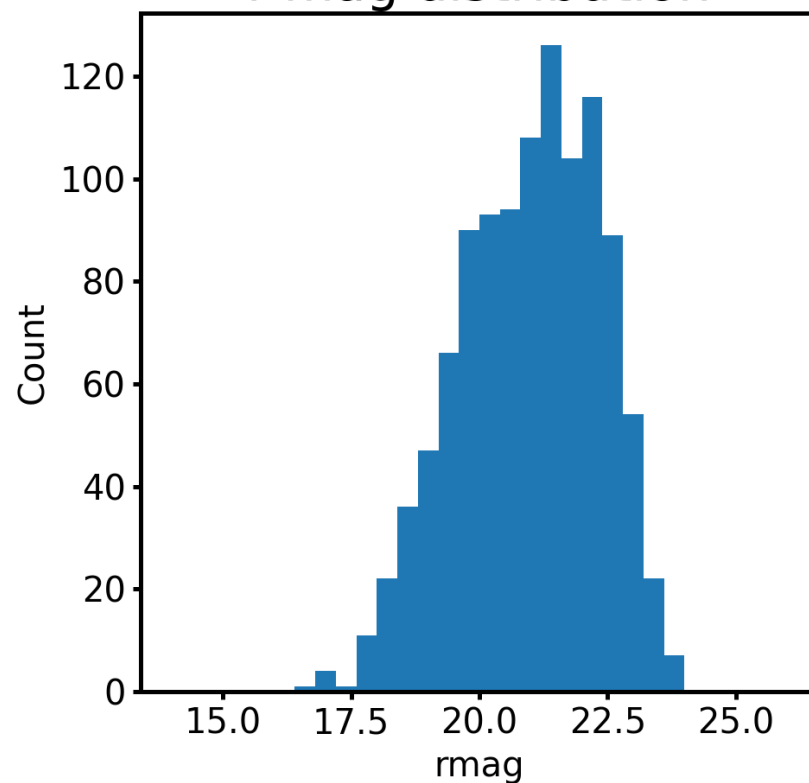
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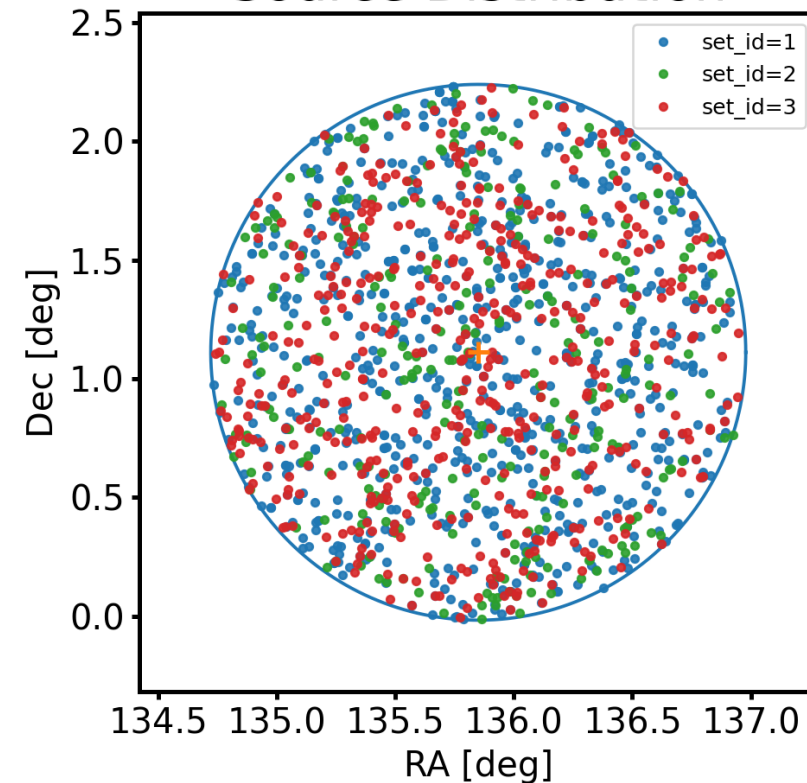
Redshift distribution

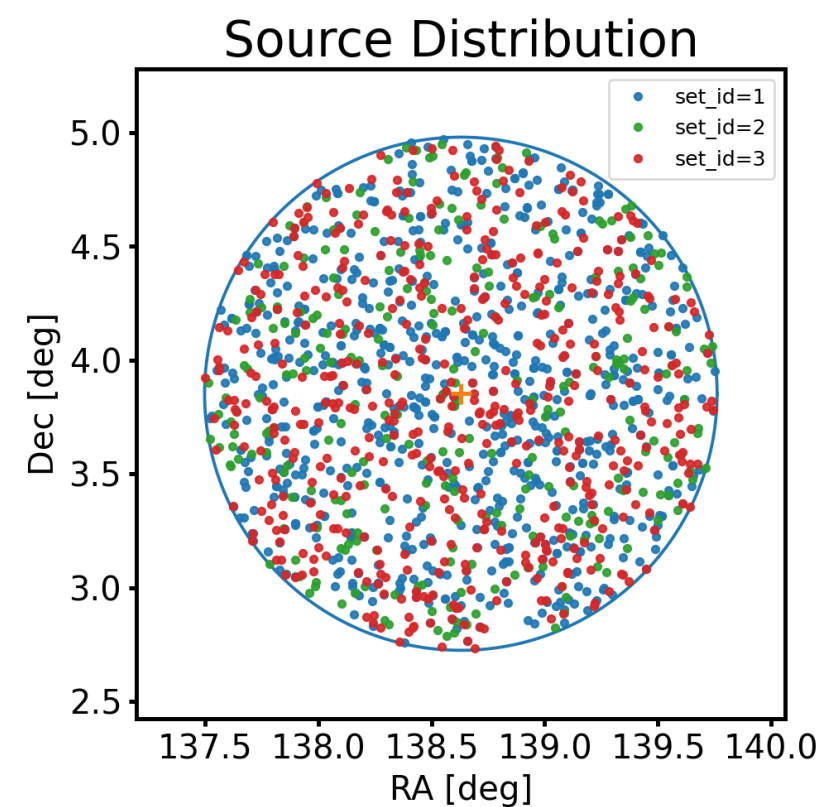
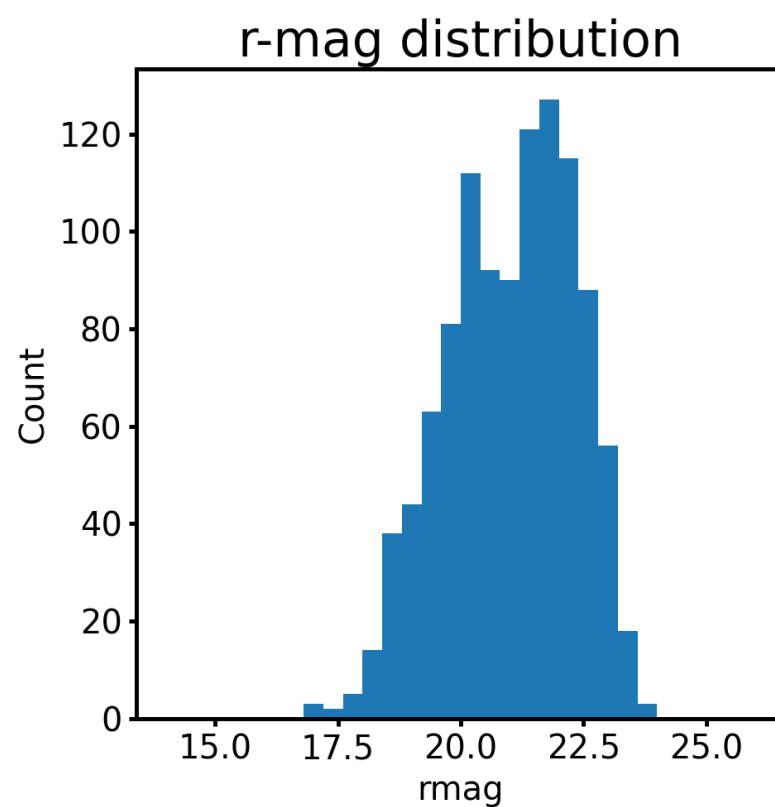
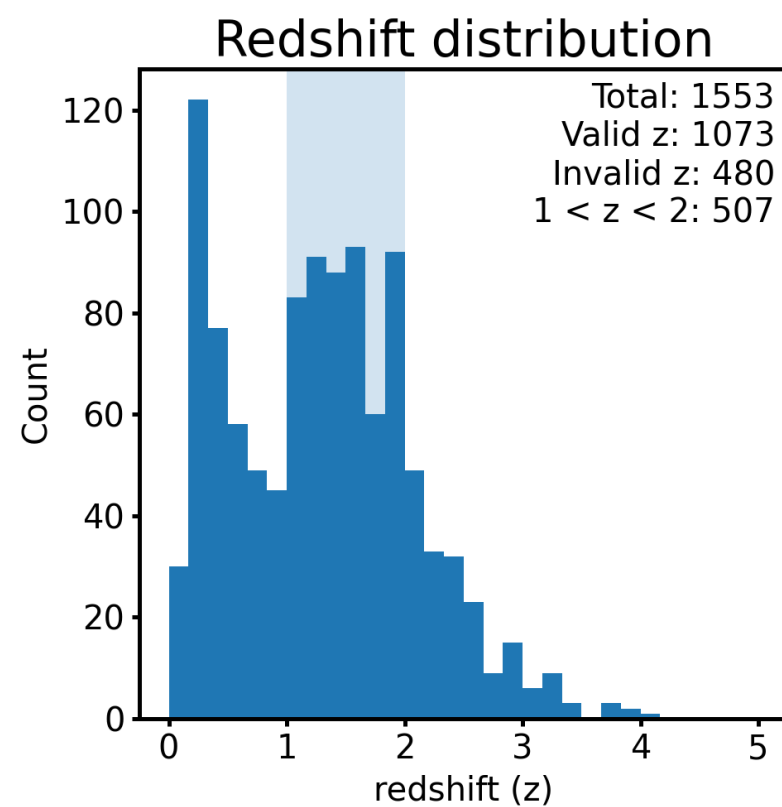
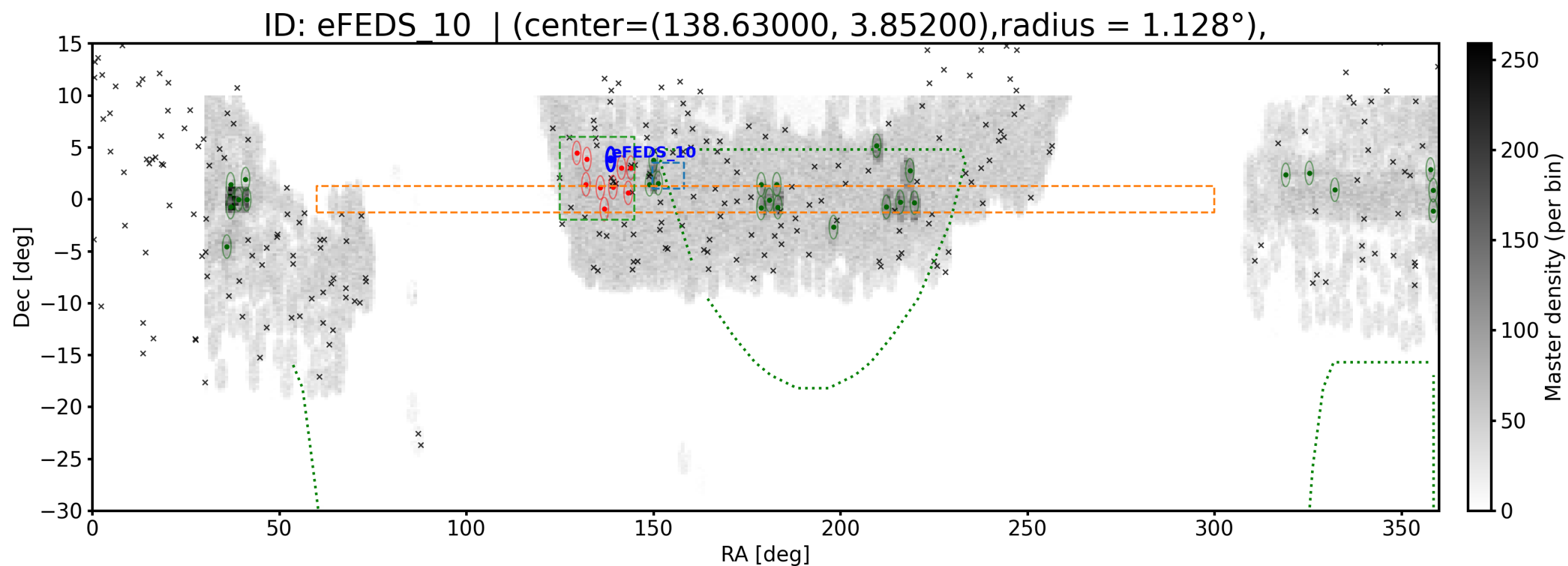


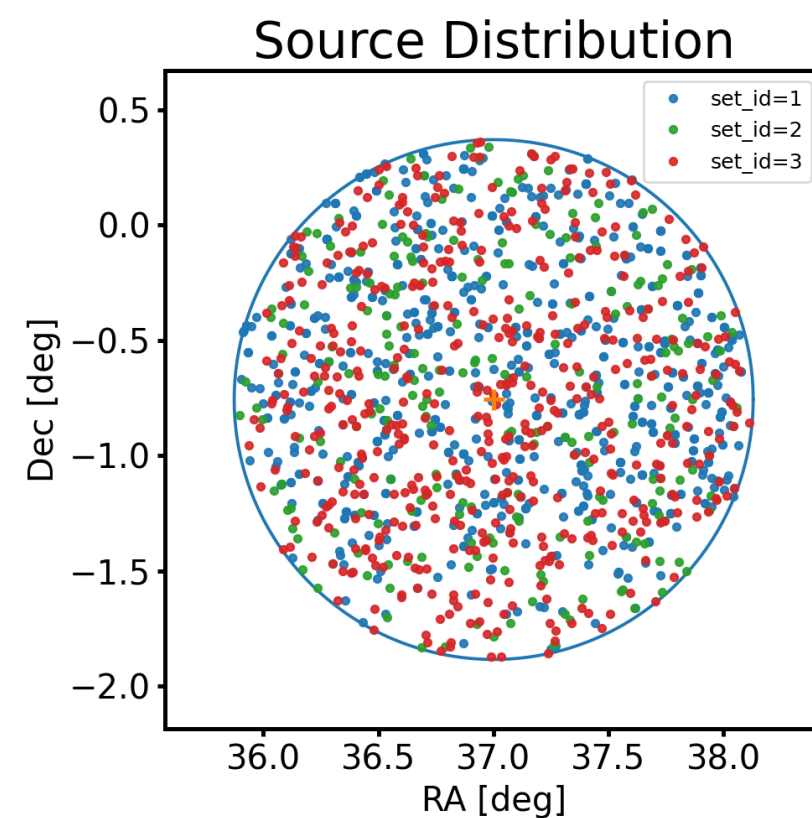
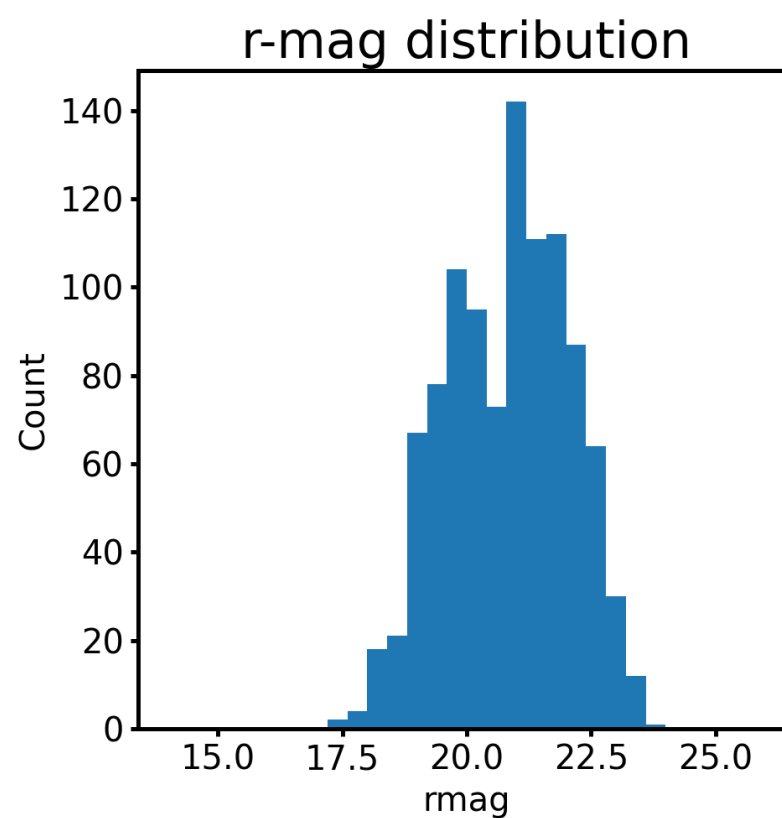
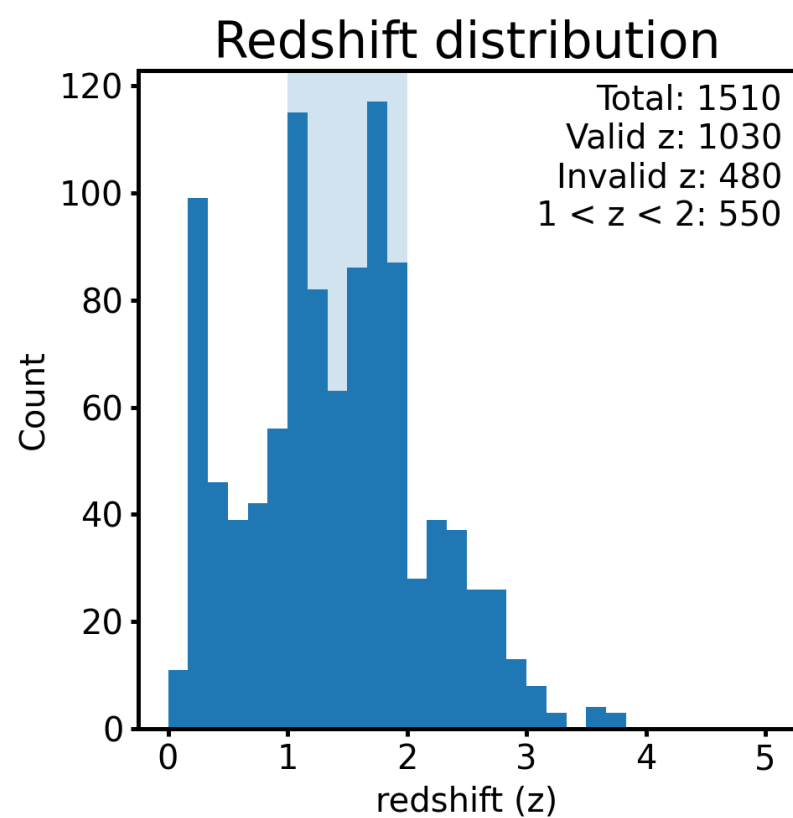
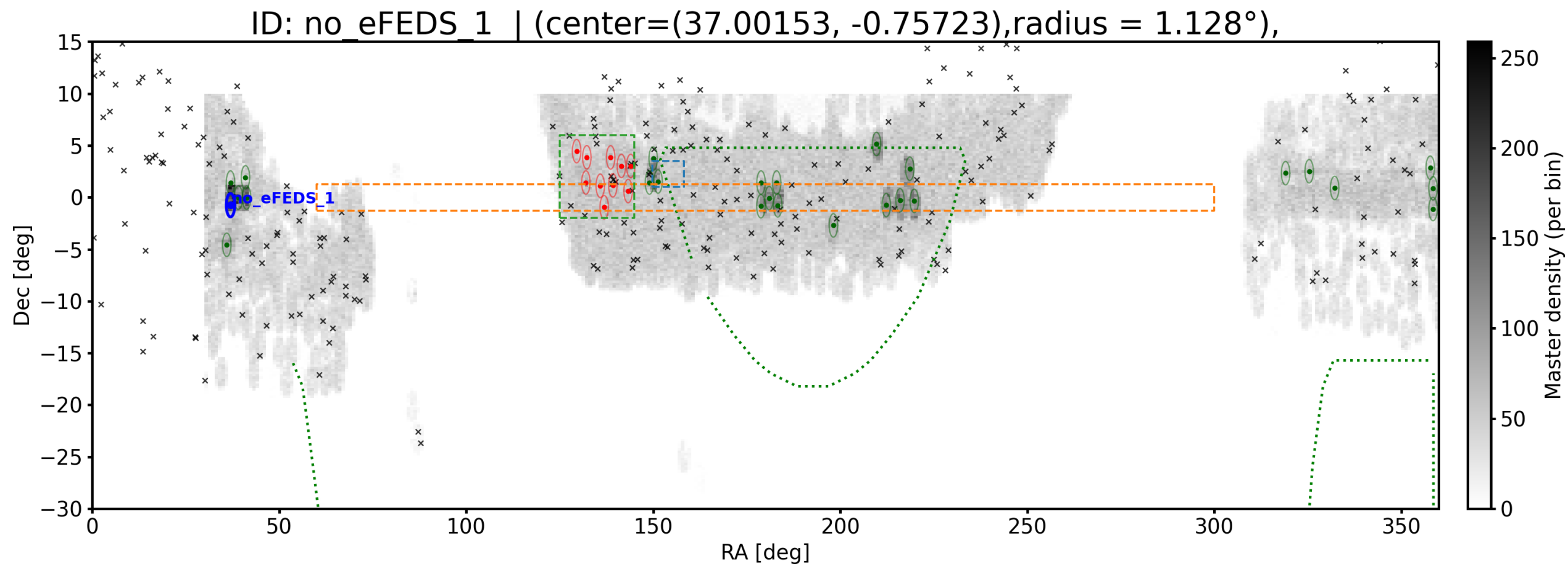
r-mag distribution

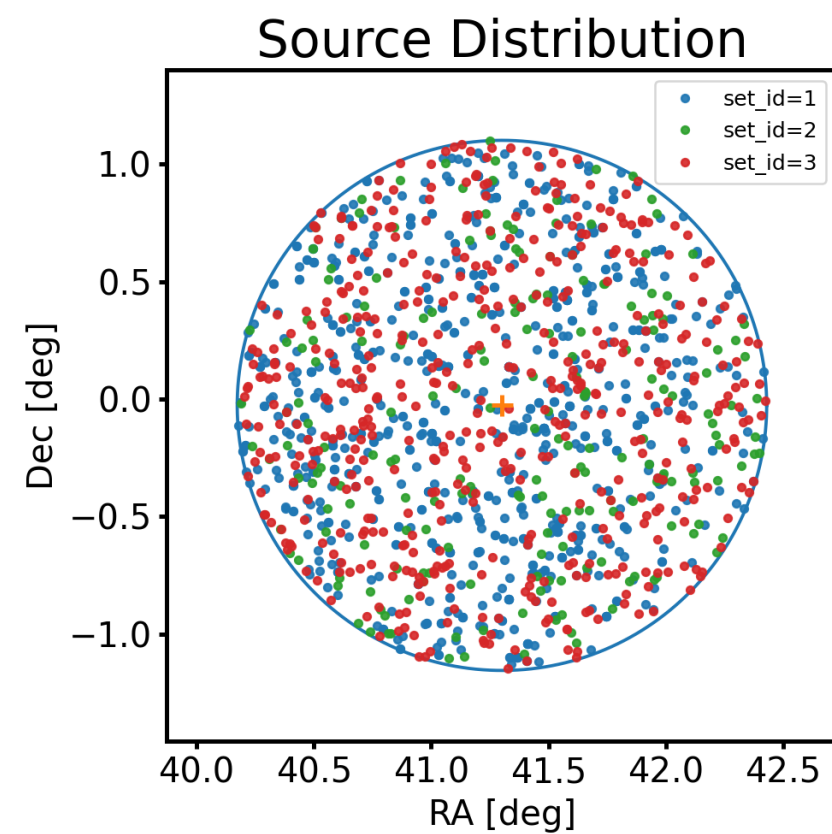
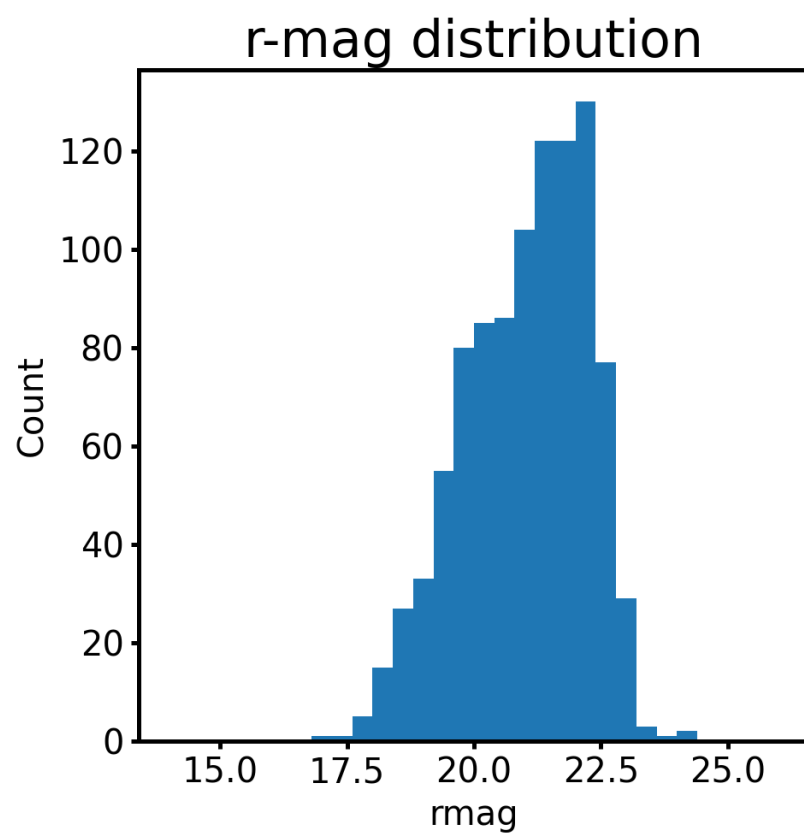
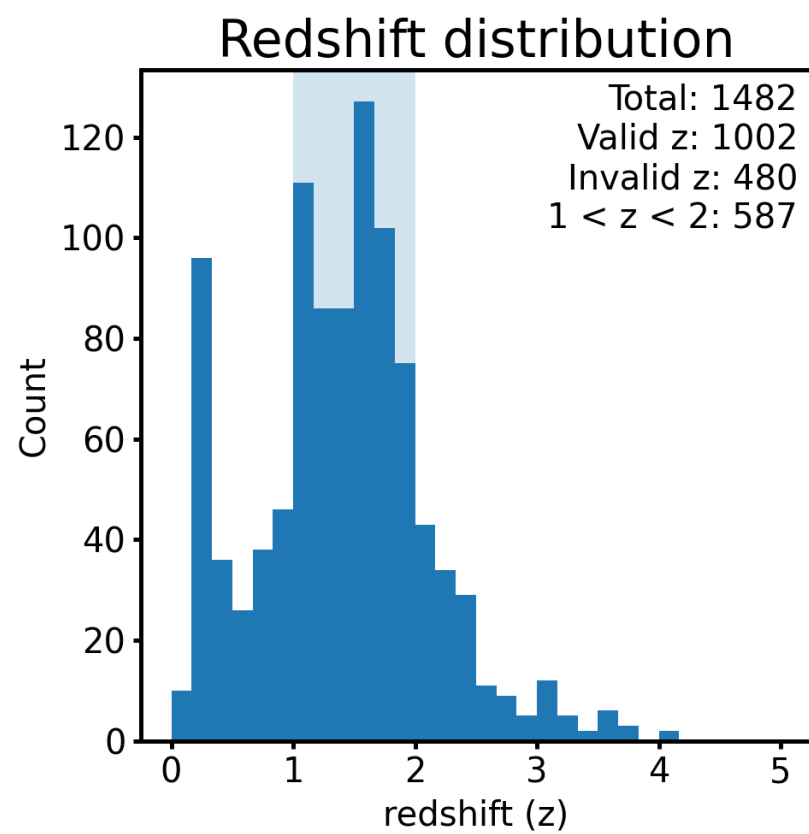
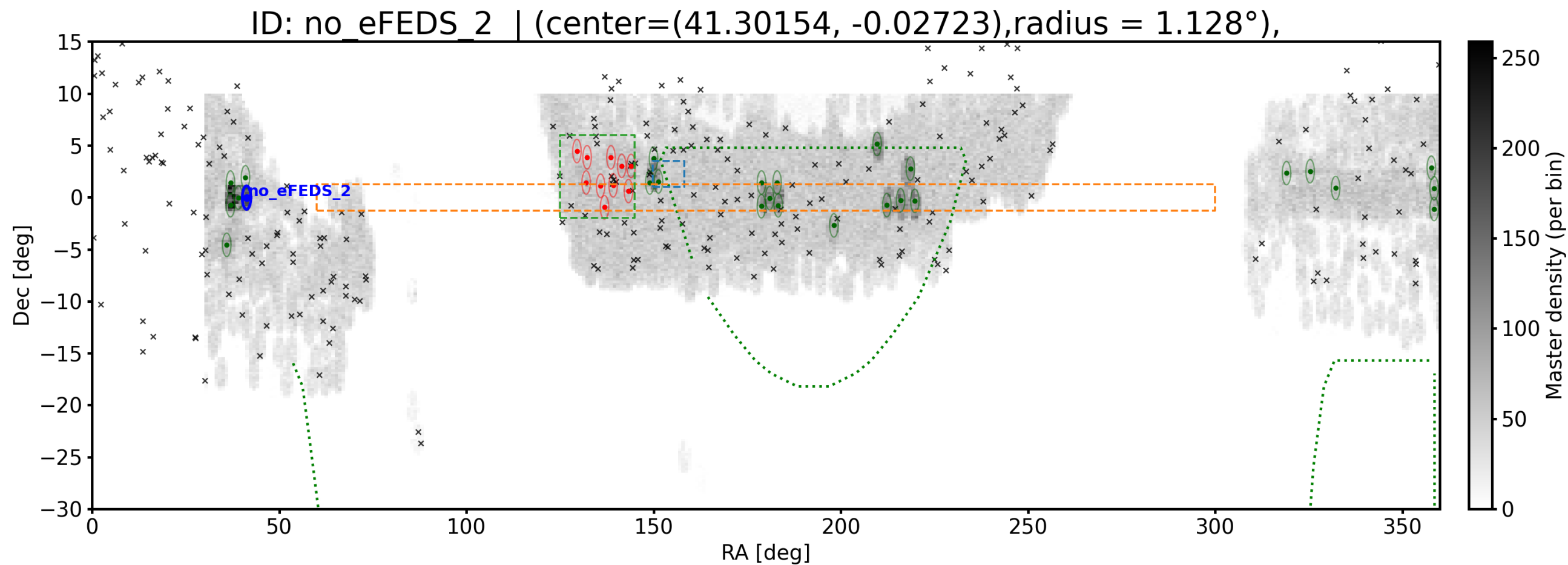


Source Distribution

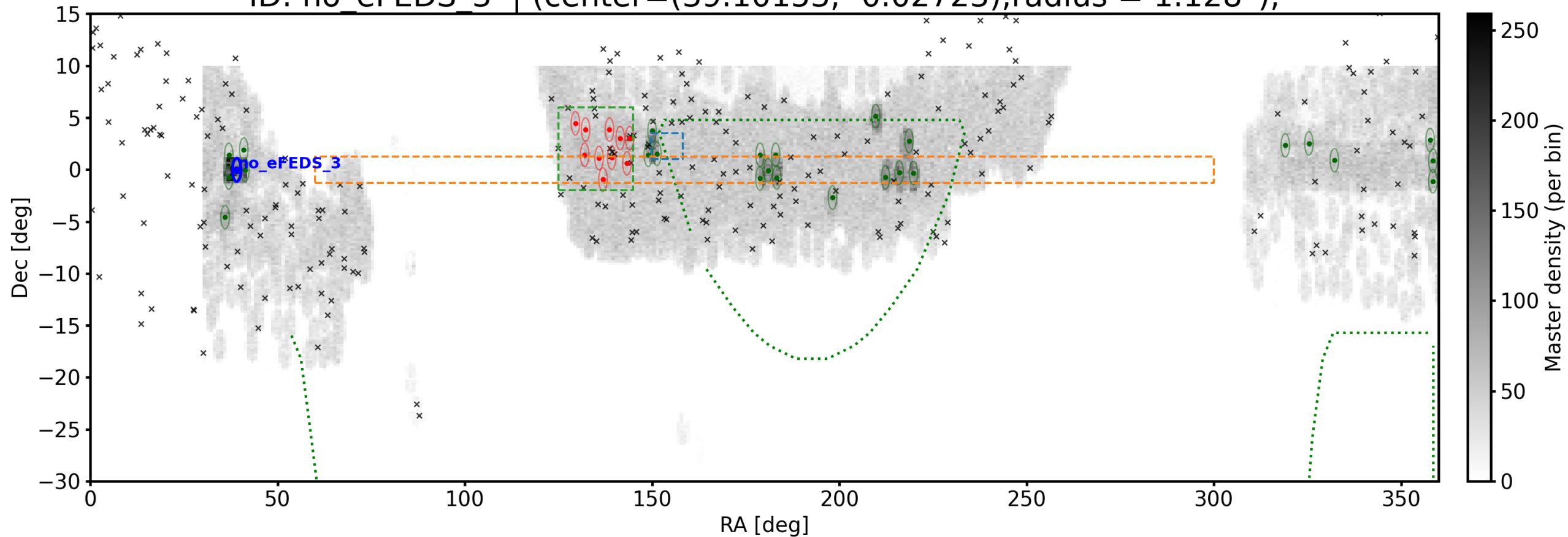




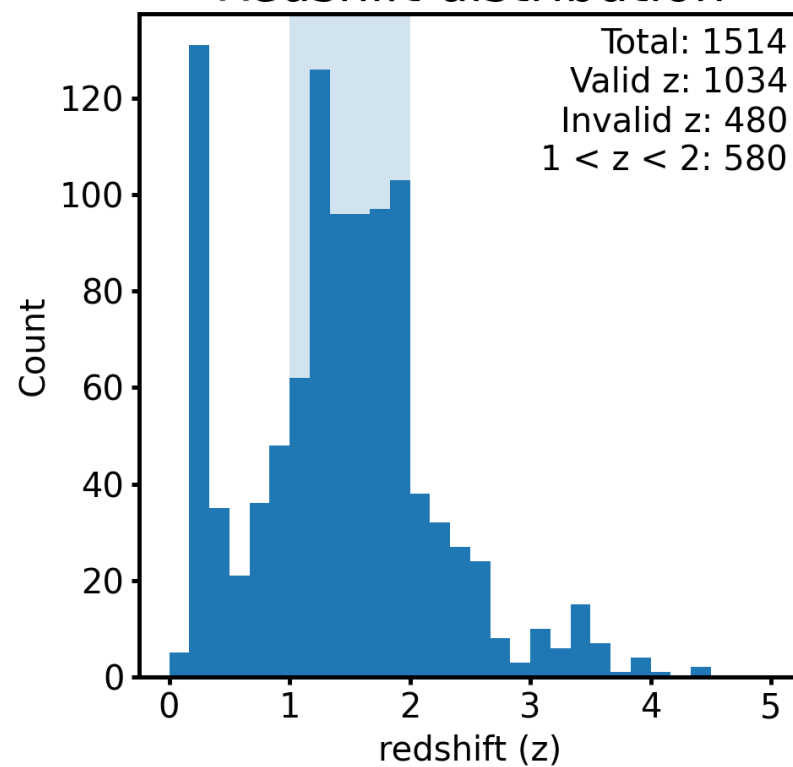




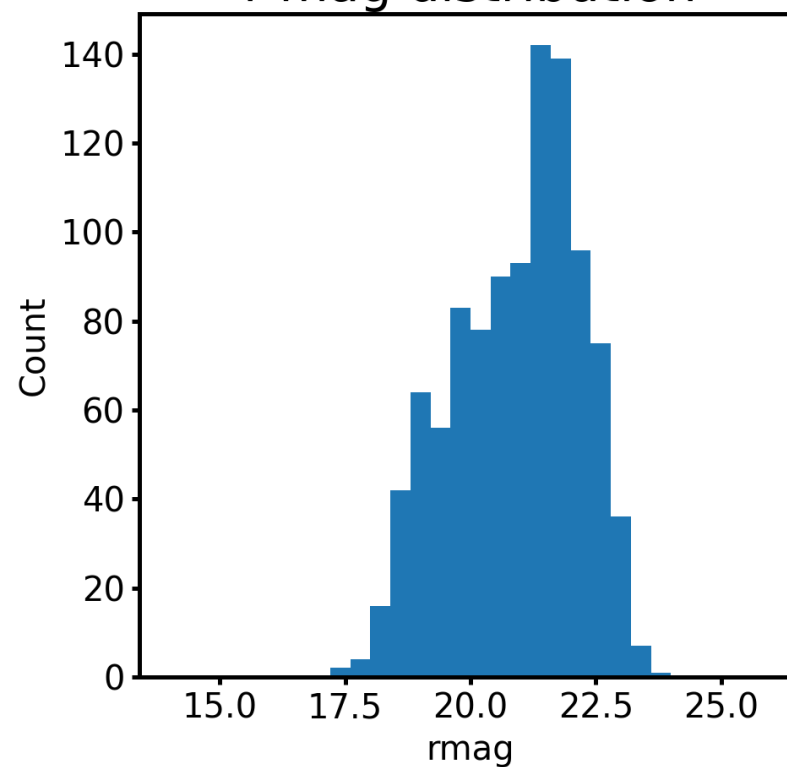
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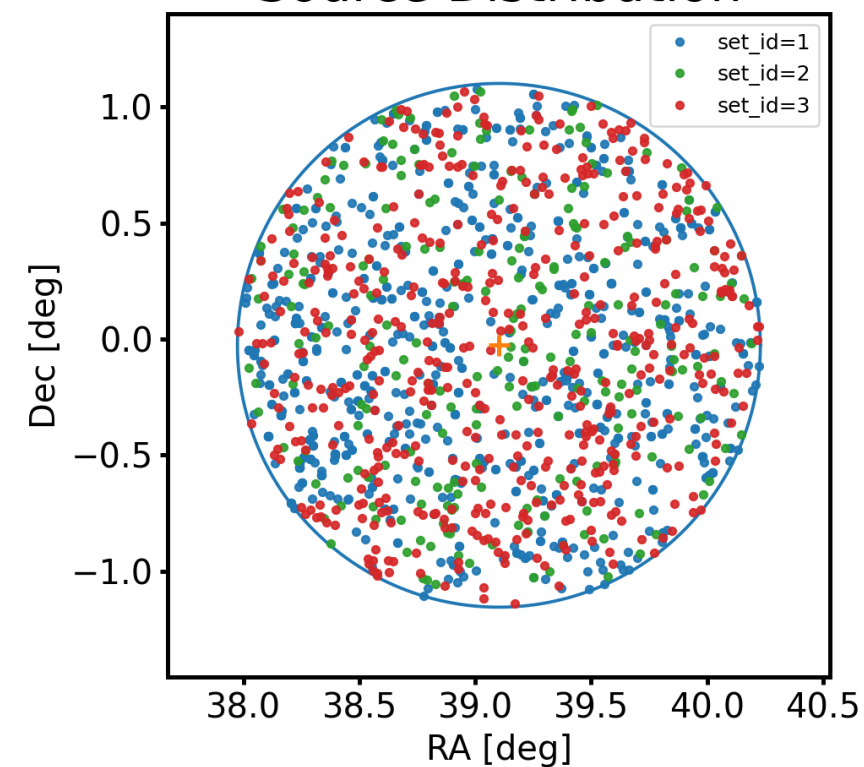
Redshift distribution



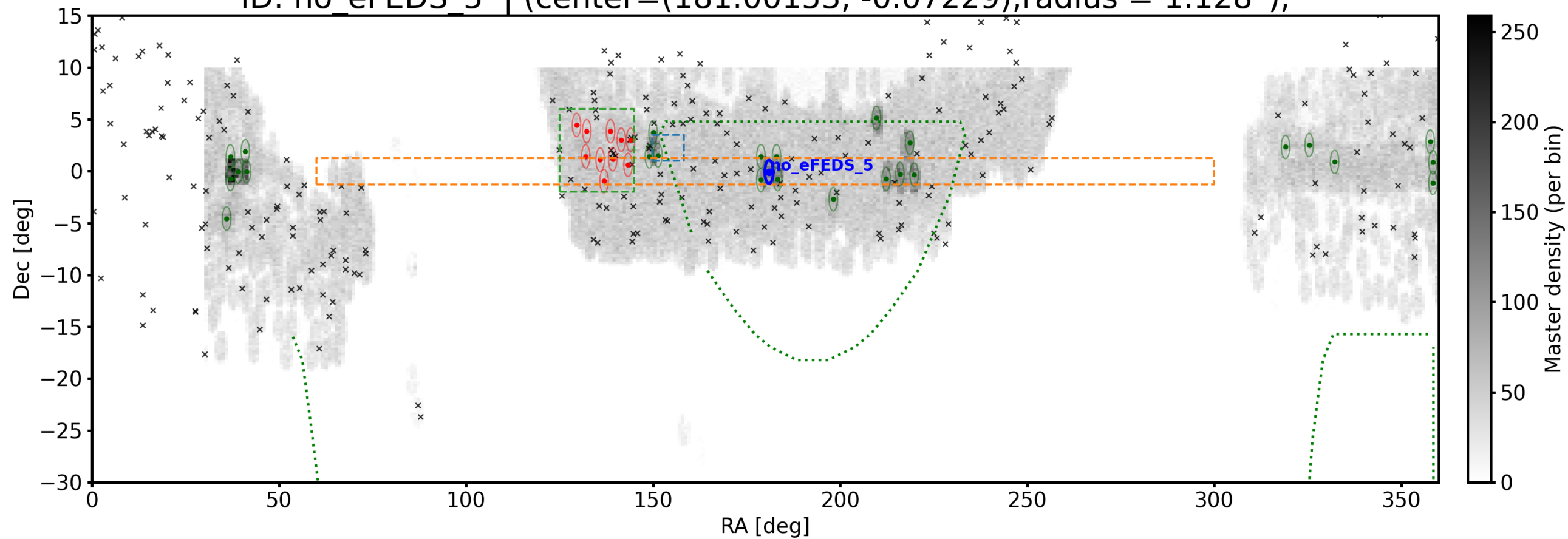
r-mag distribution



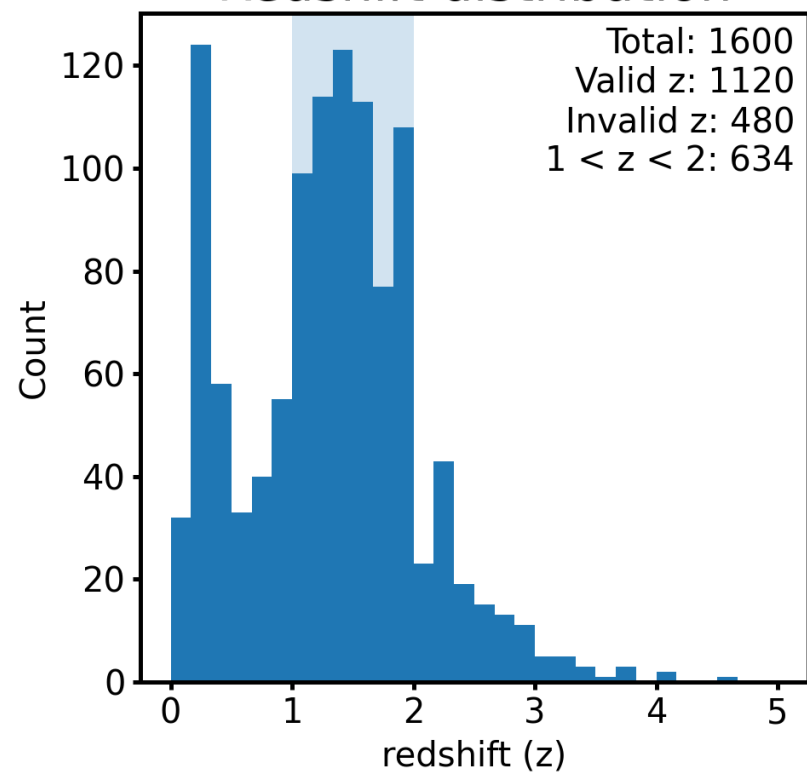
Source Distribution



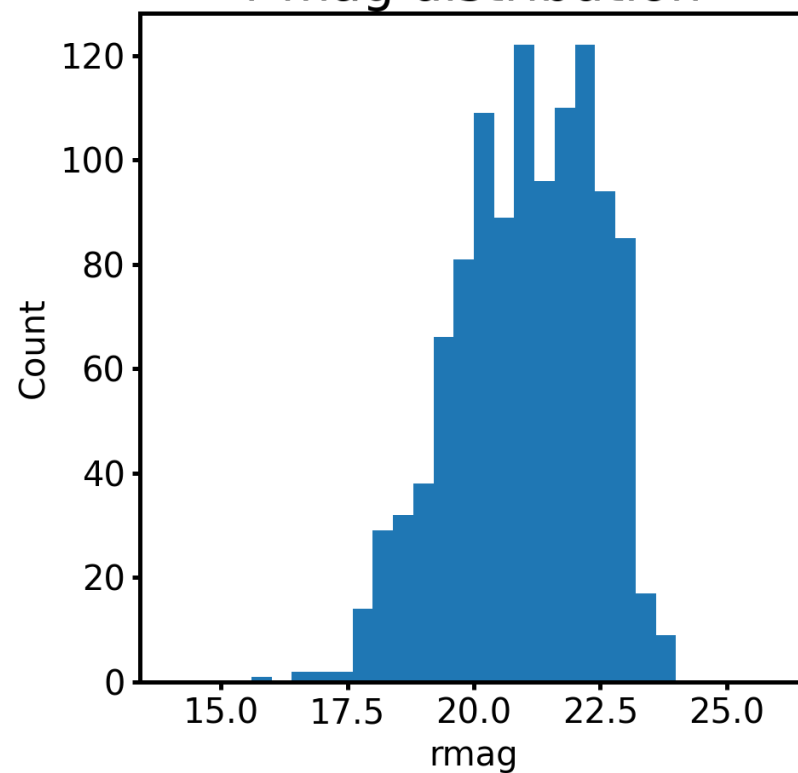
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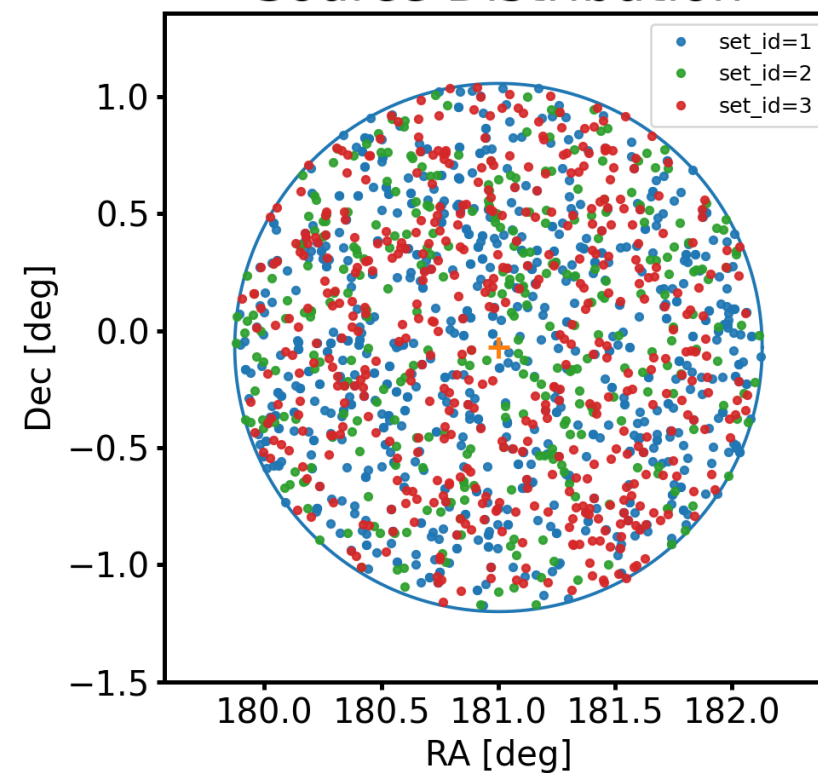
Redshift distribution

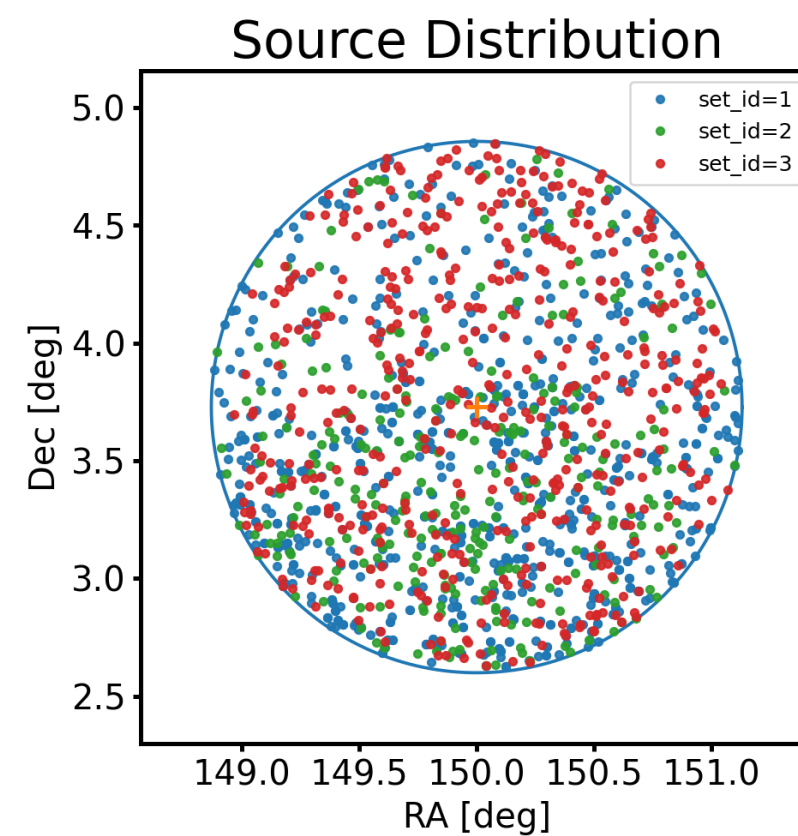
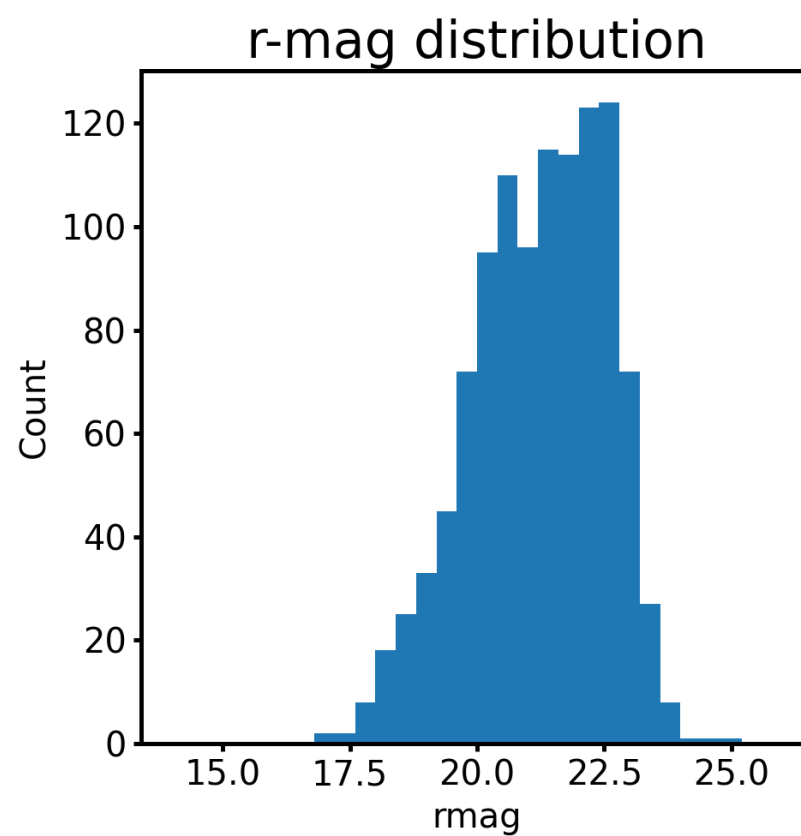
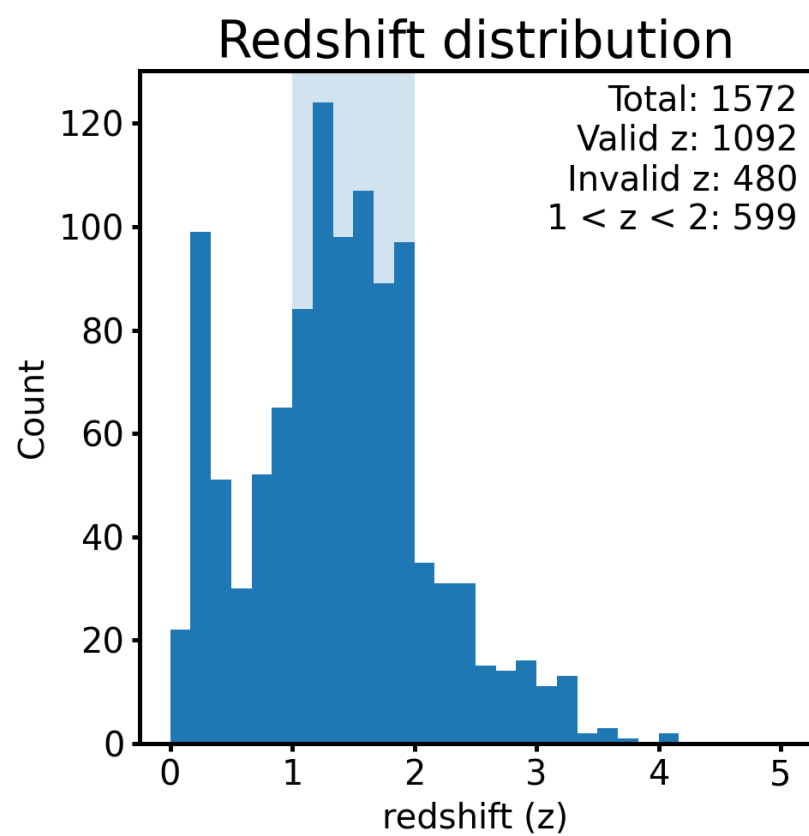
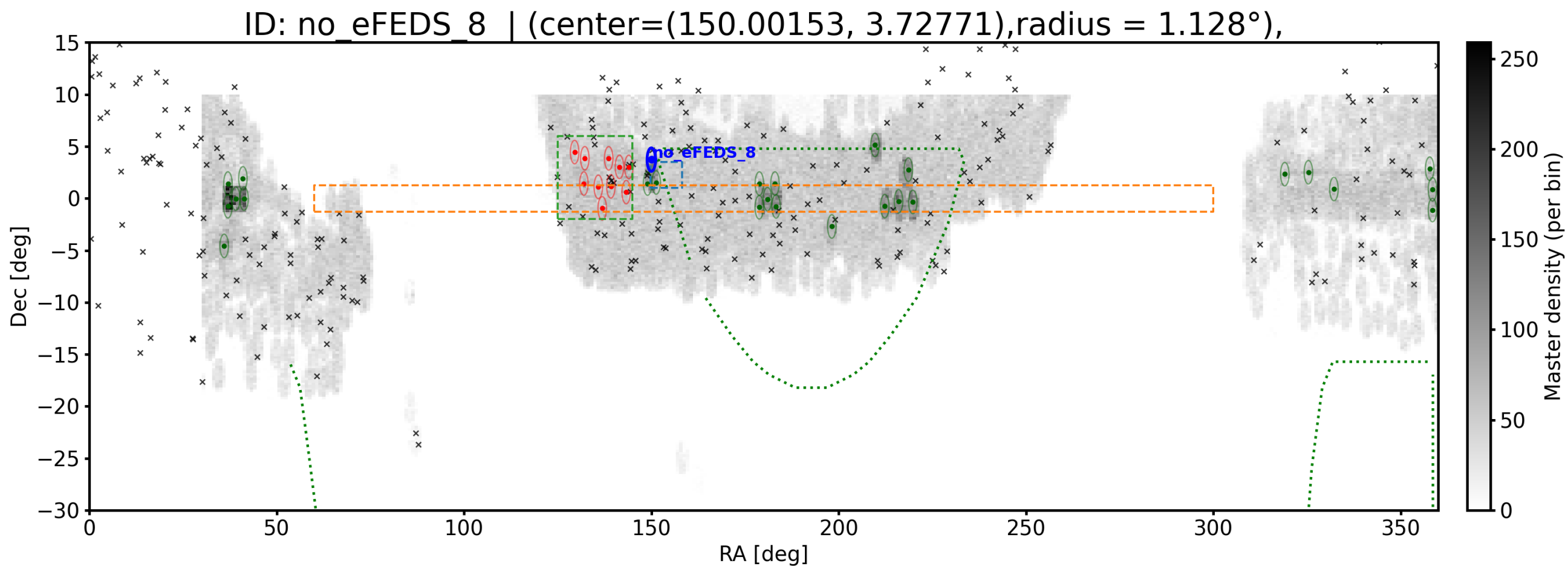


r-mag distribution

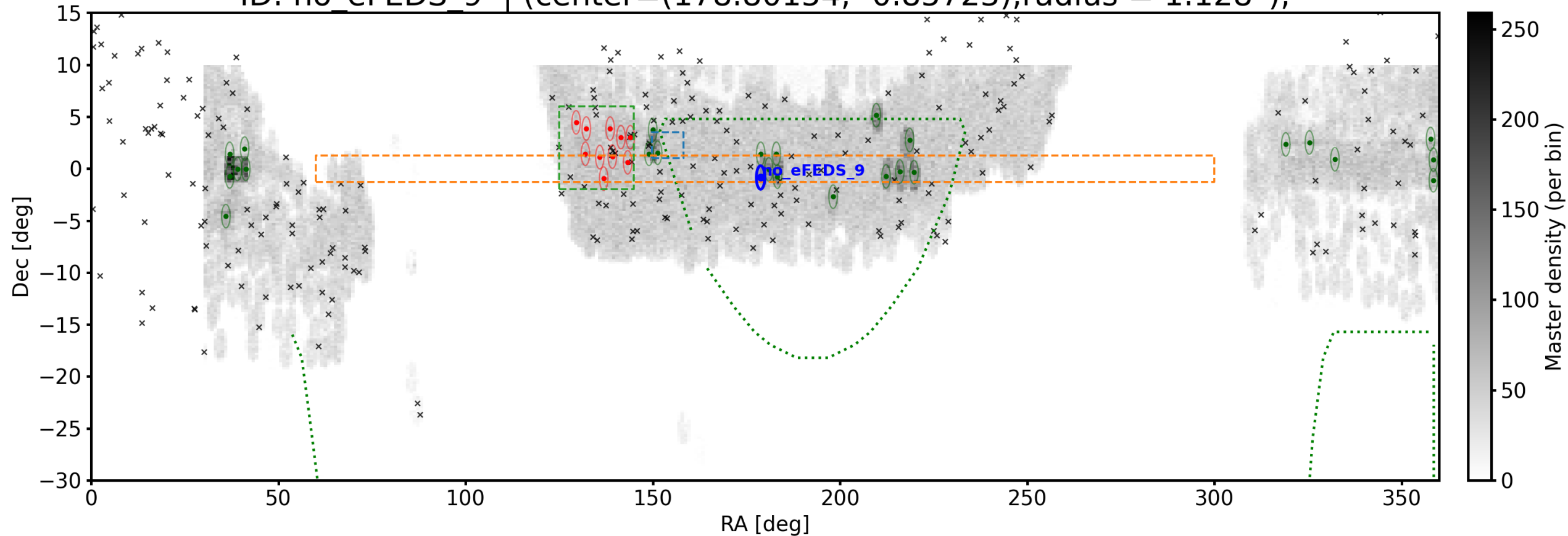


Source Distribution

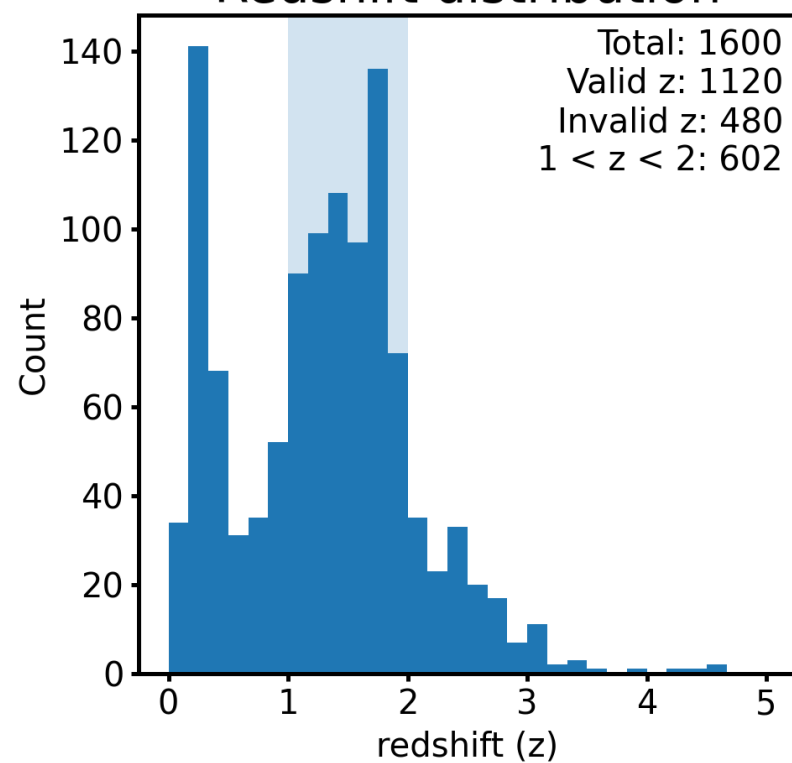




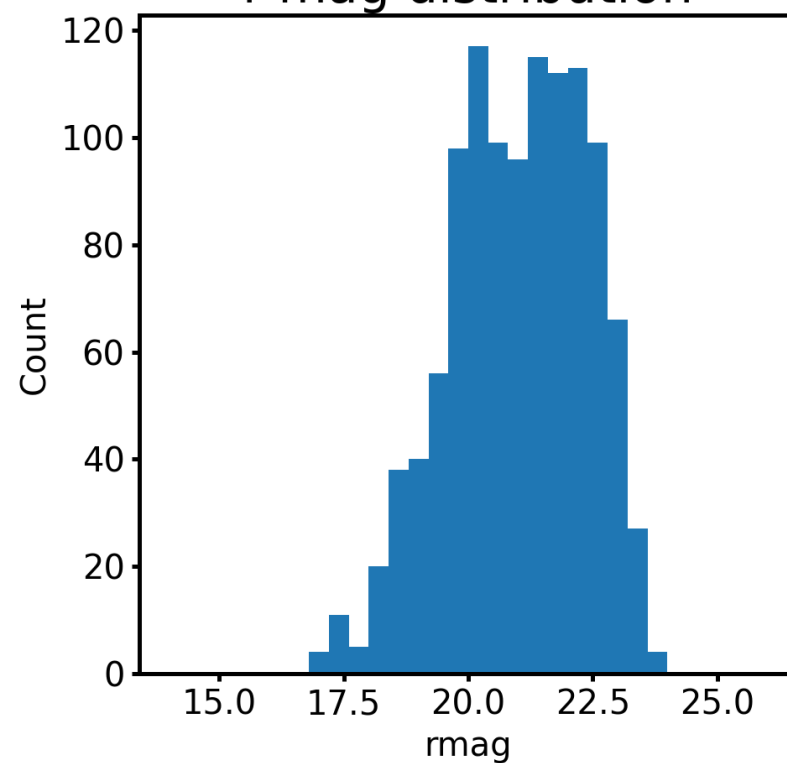
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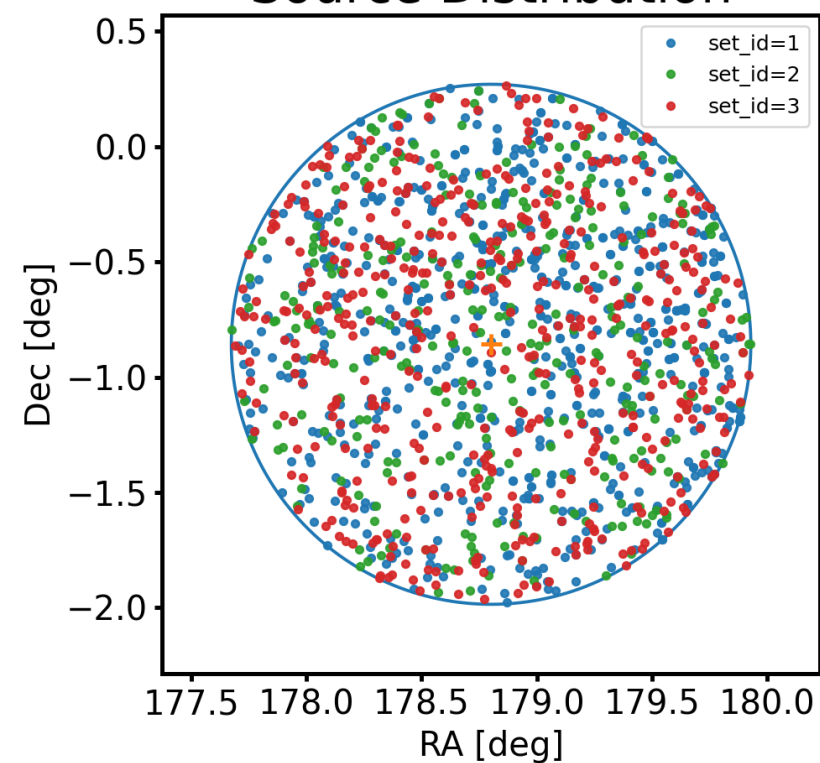
Redshift distribution

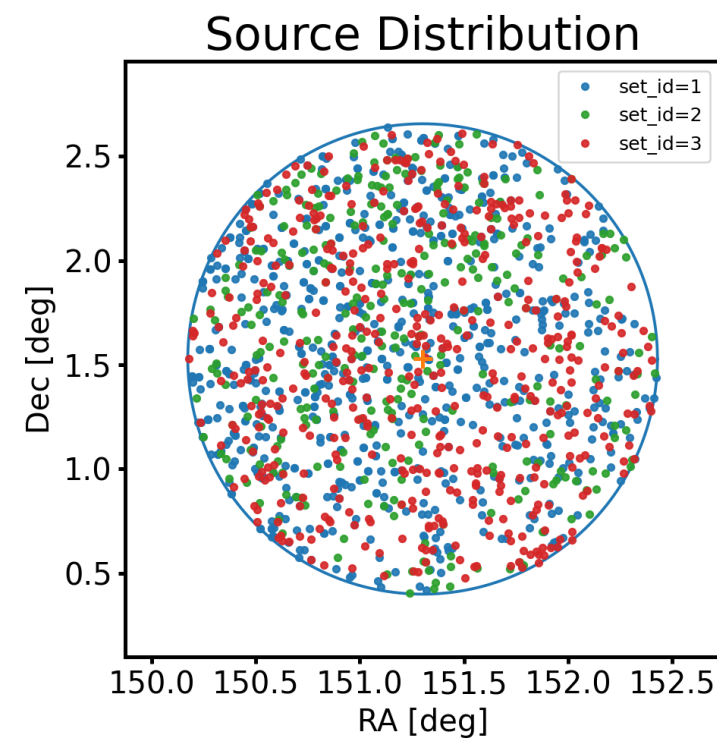
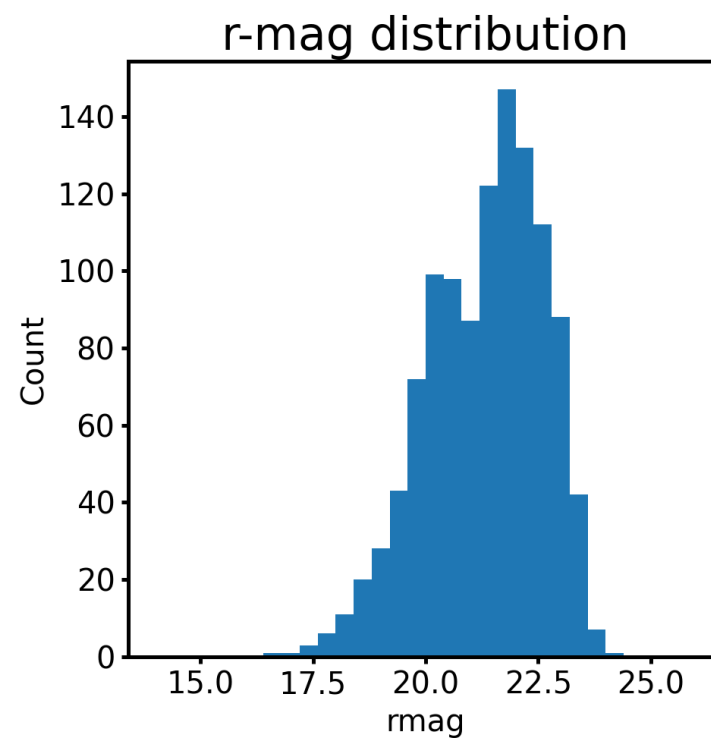
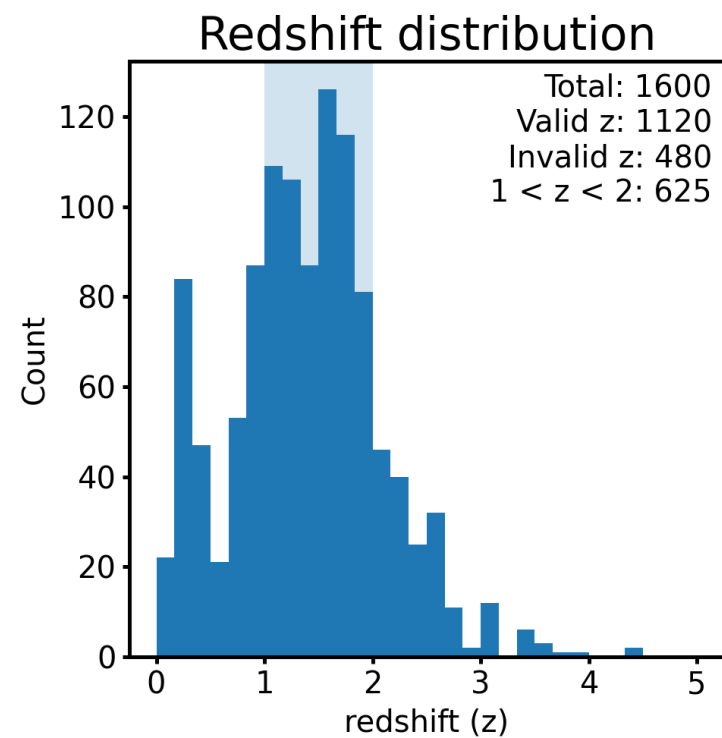
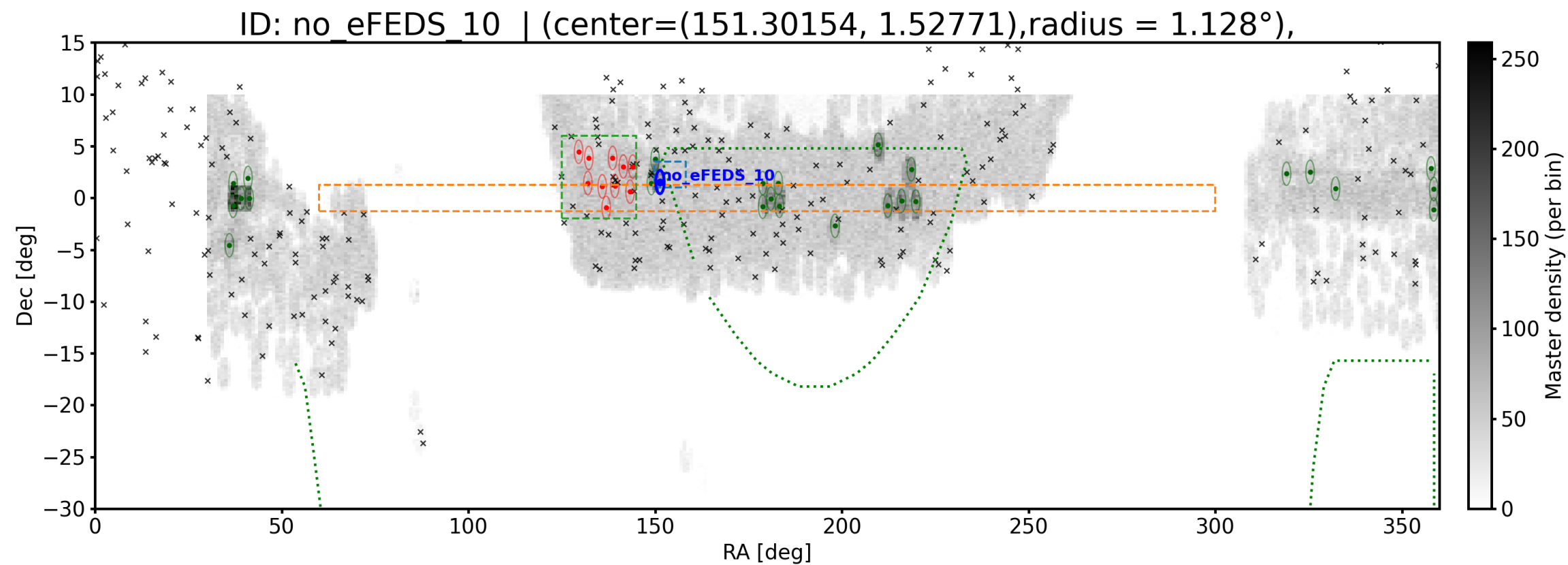


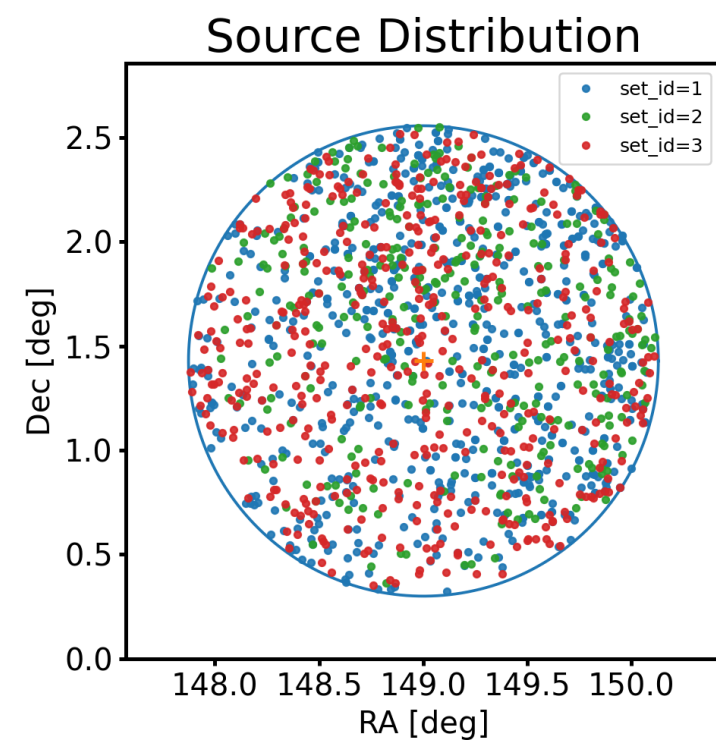
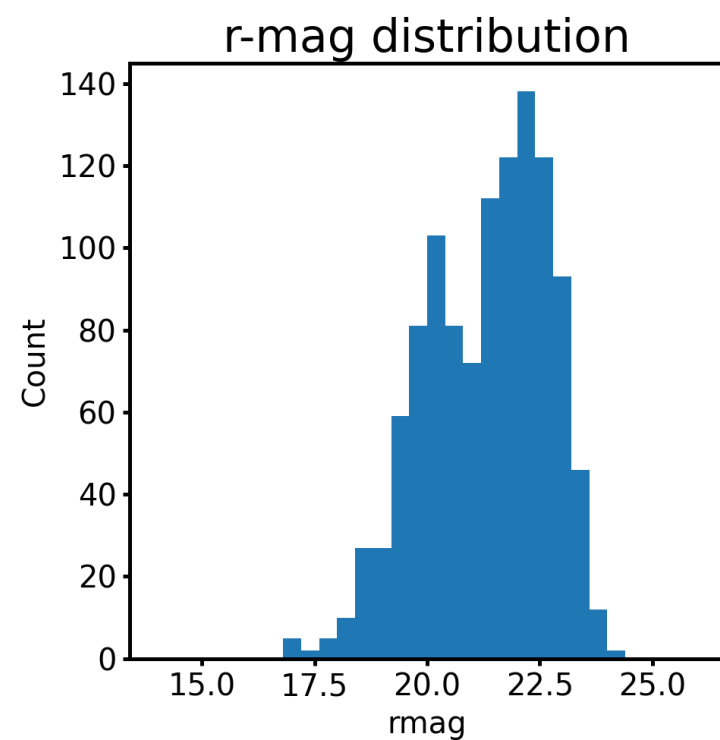
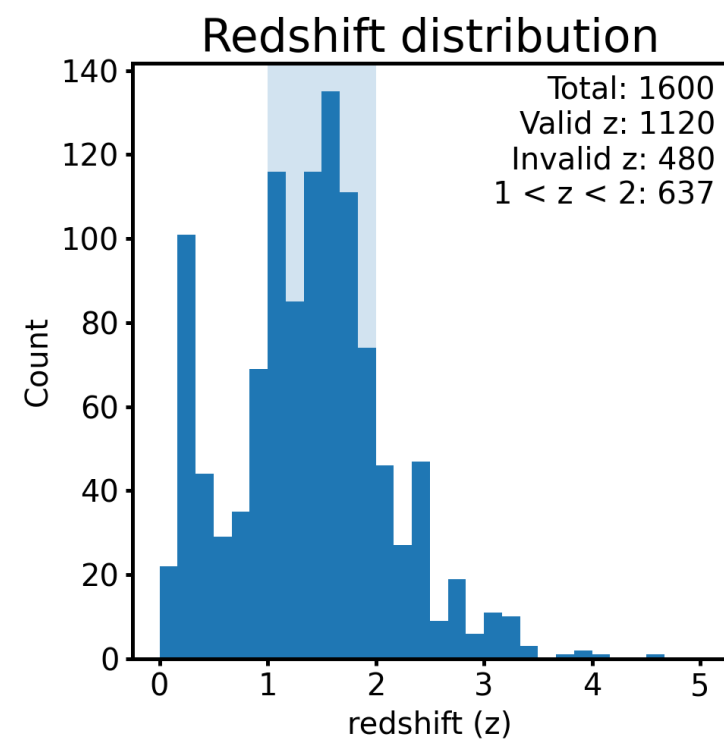
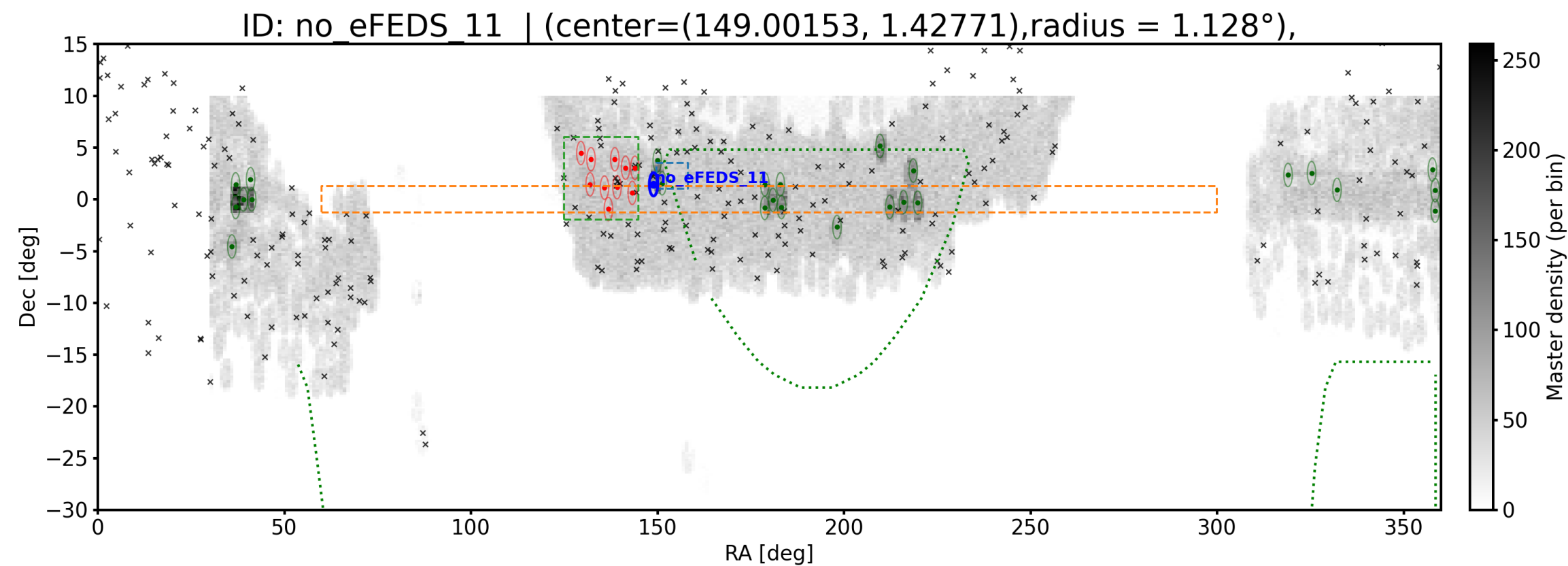
r-mag distribution

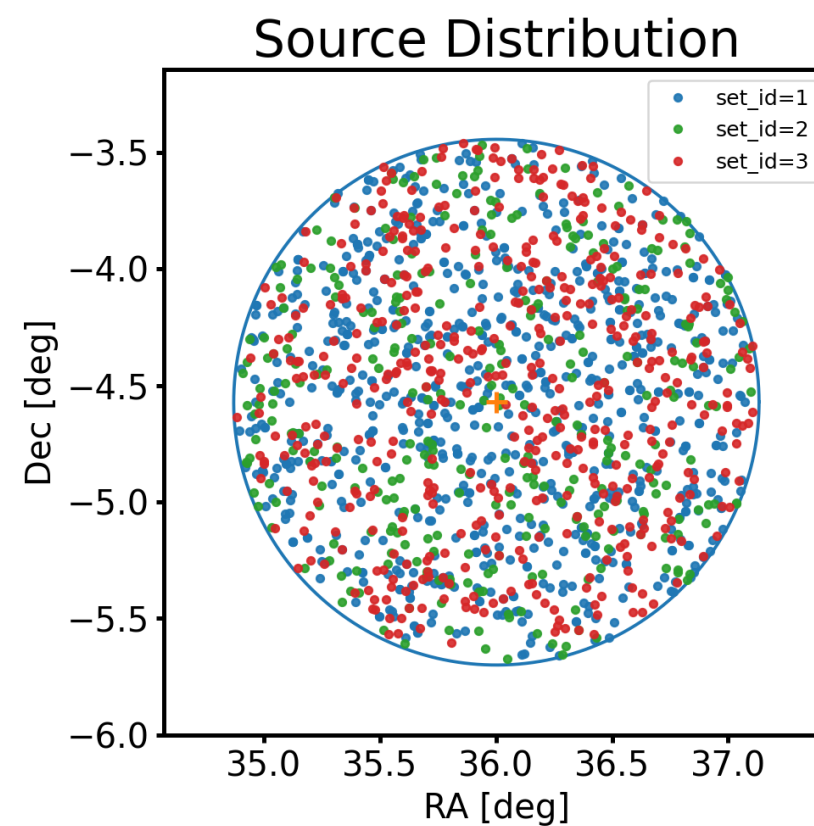
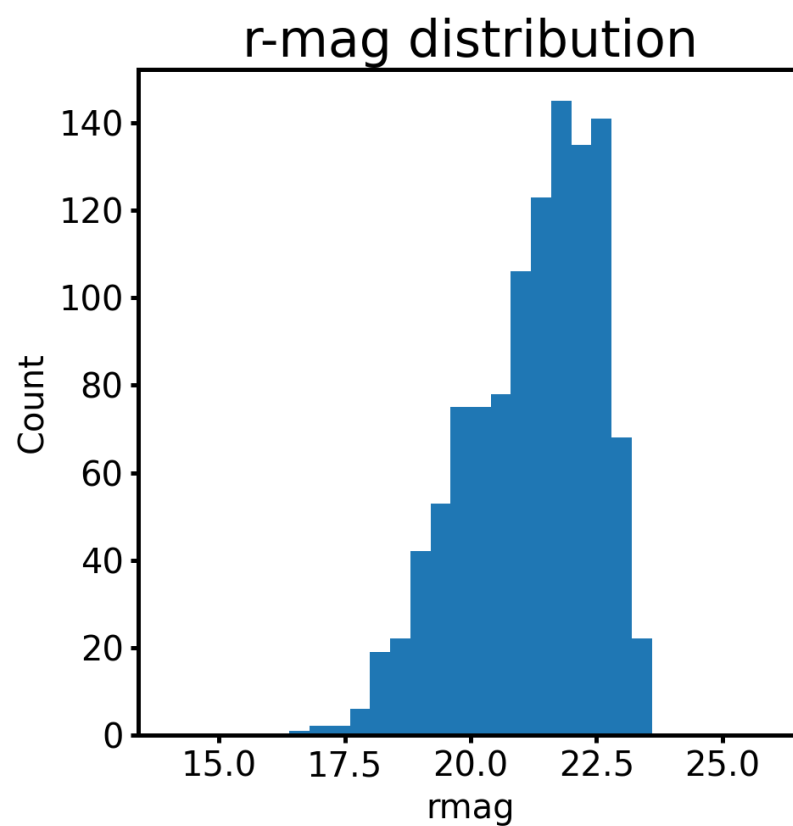
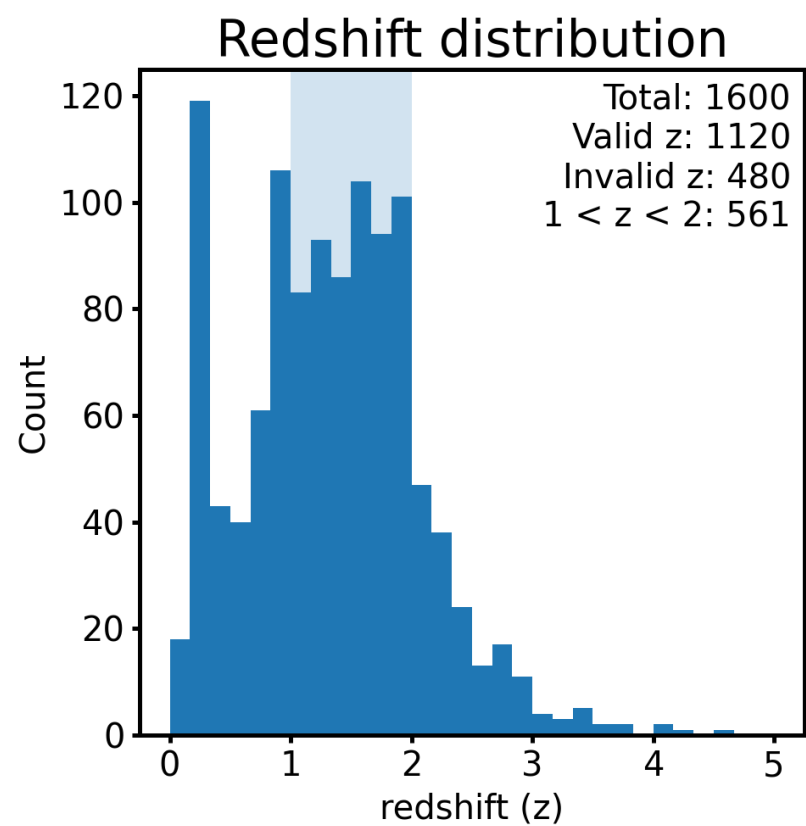
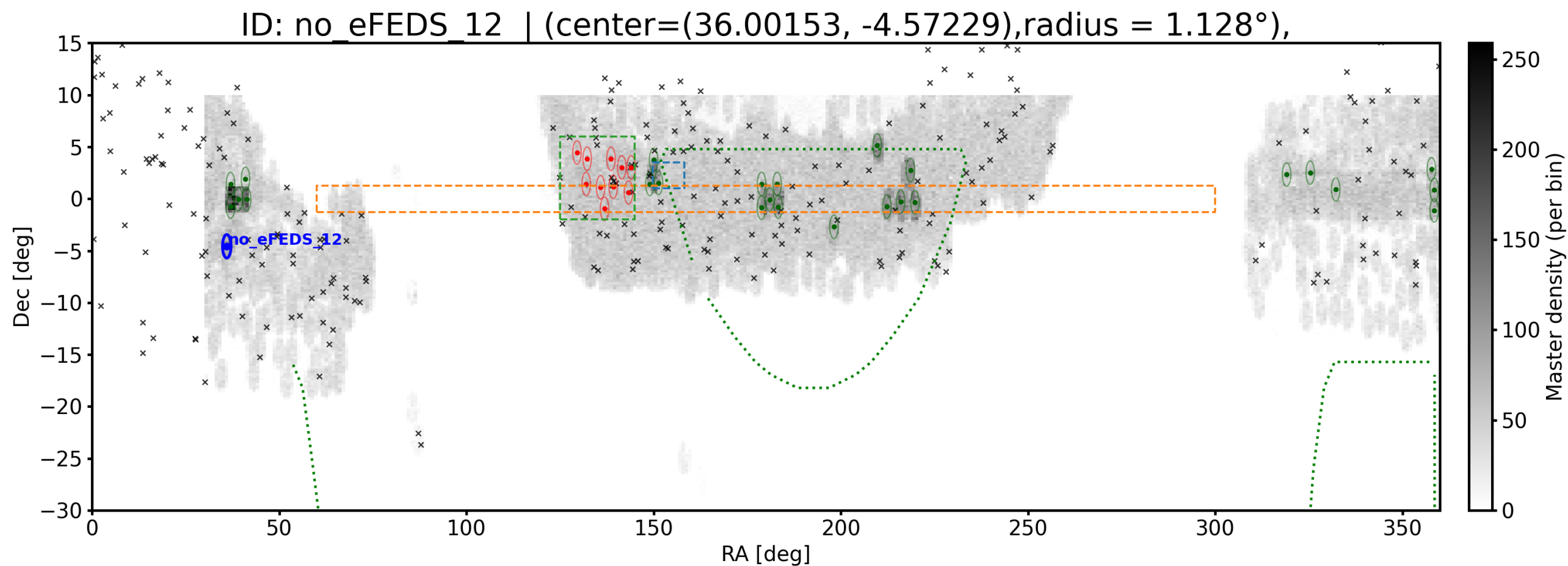


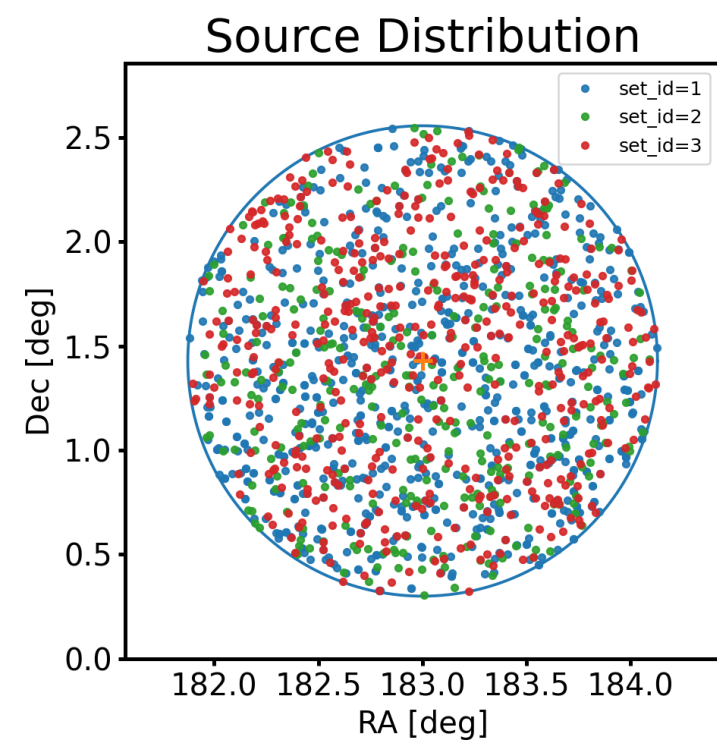
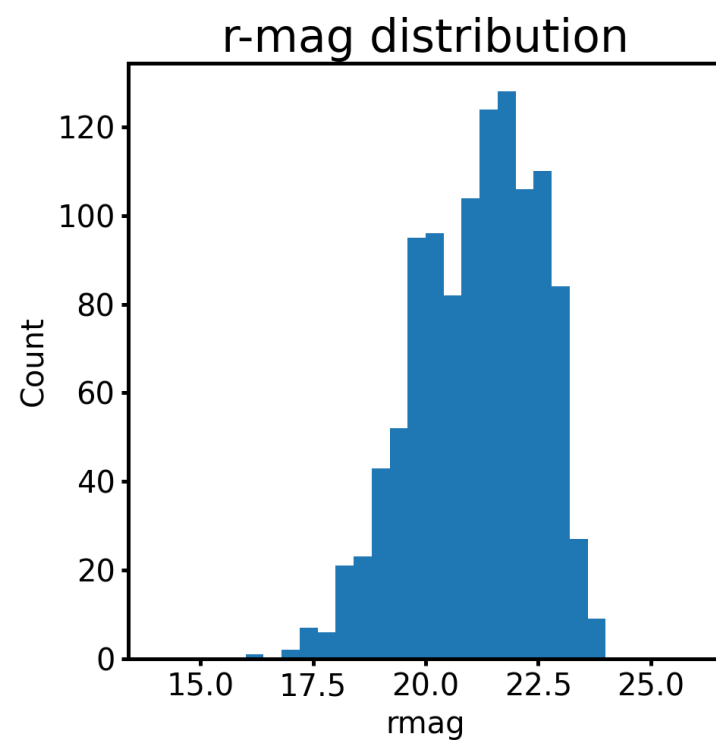
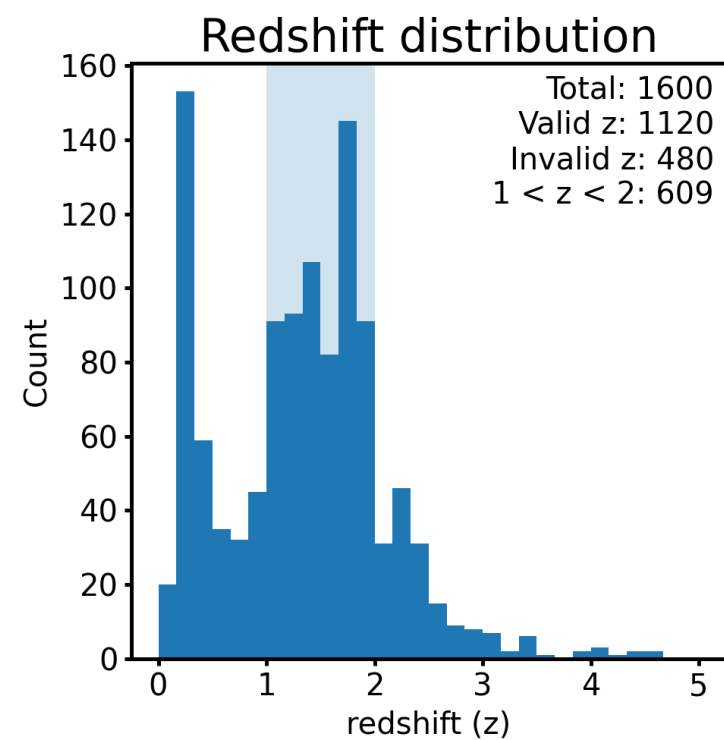
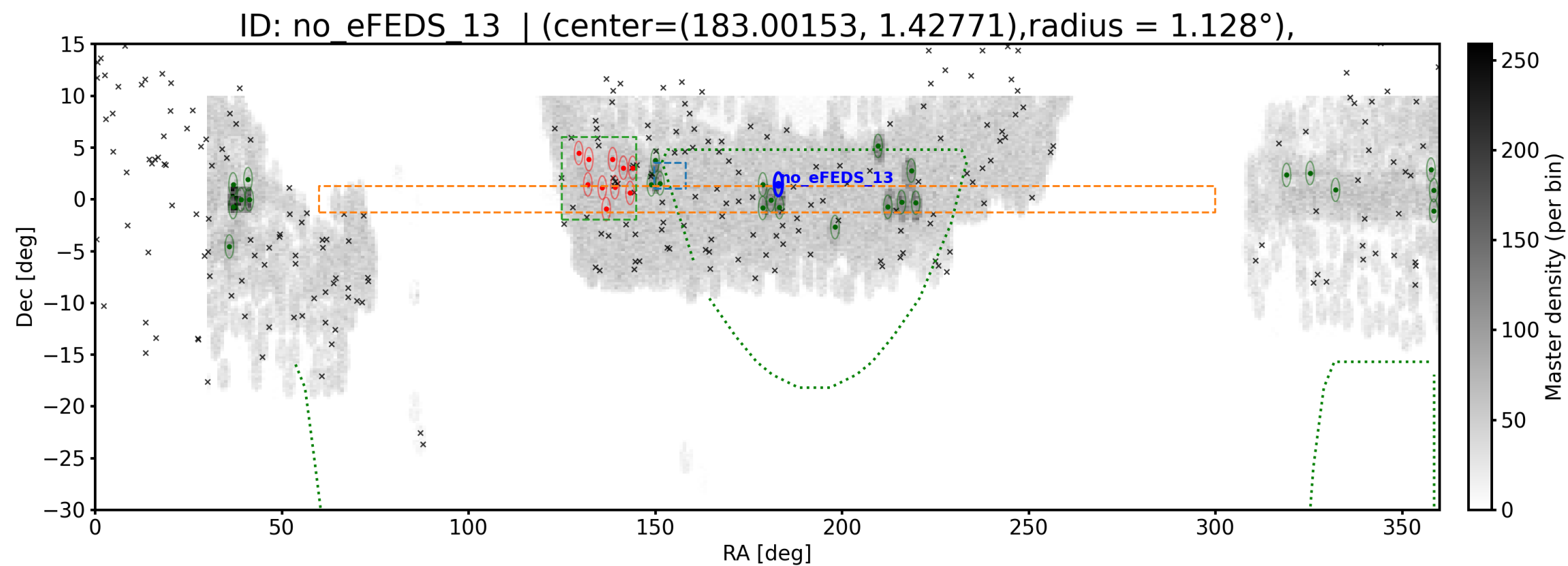
Source Distribution

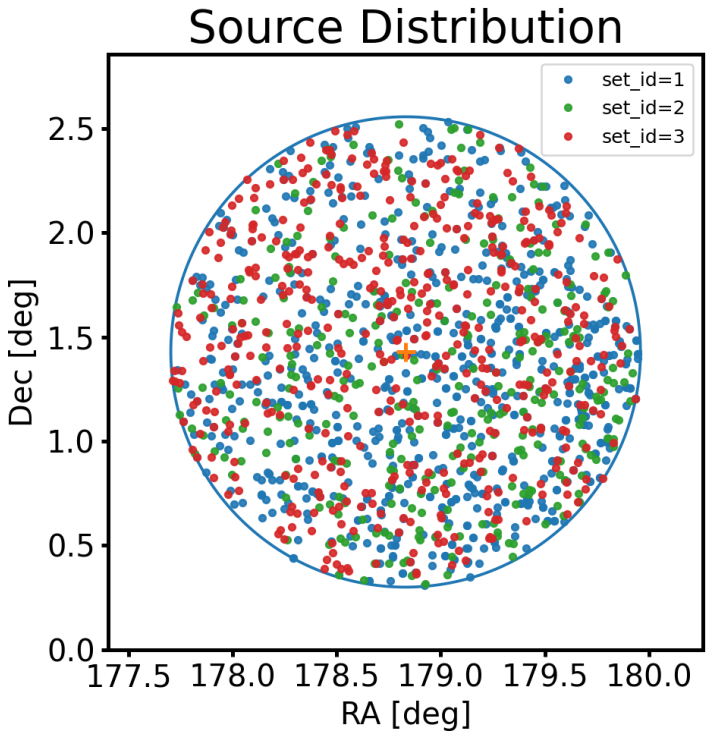
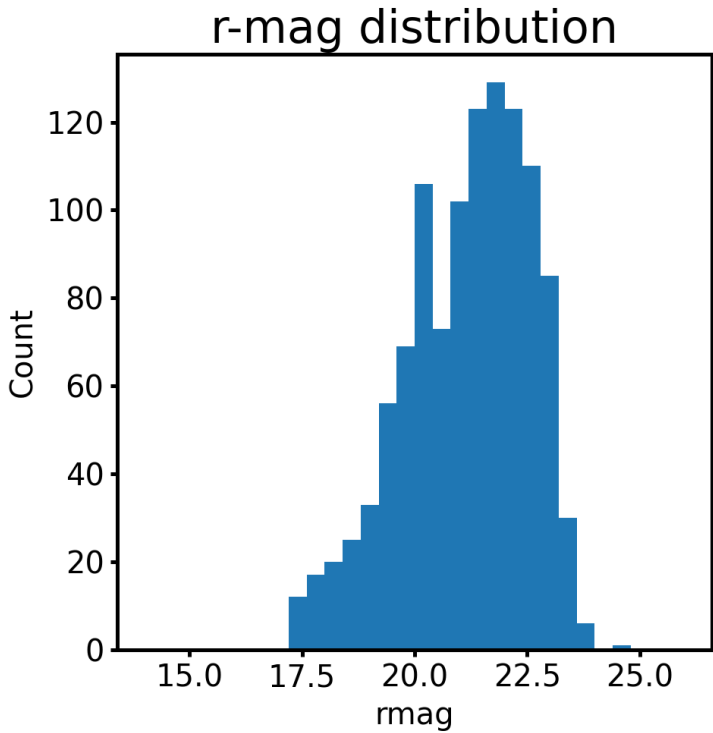
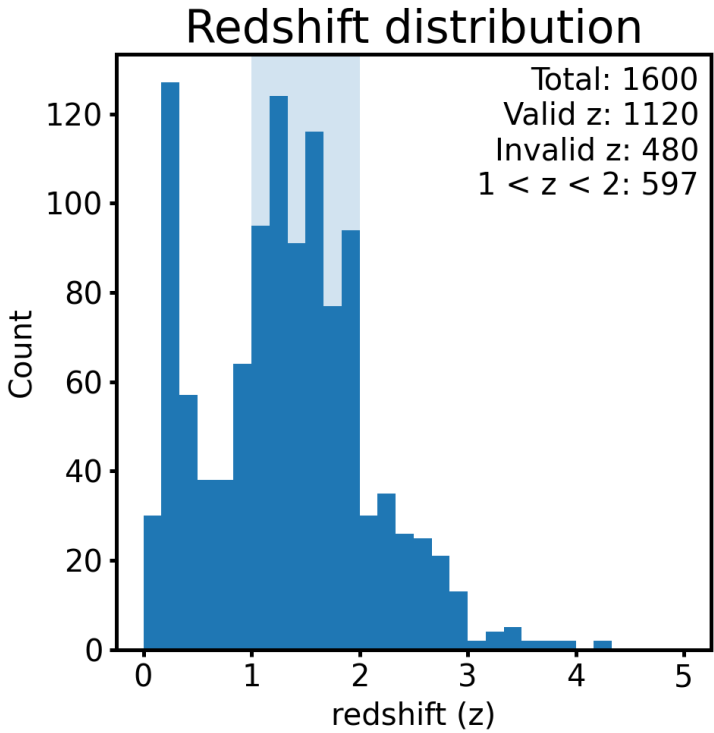
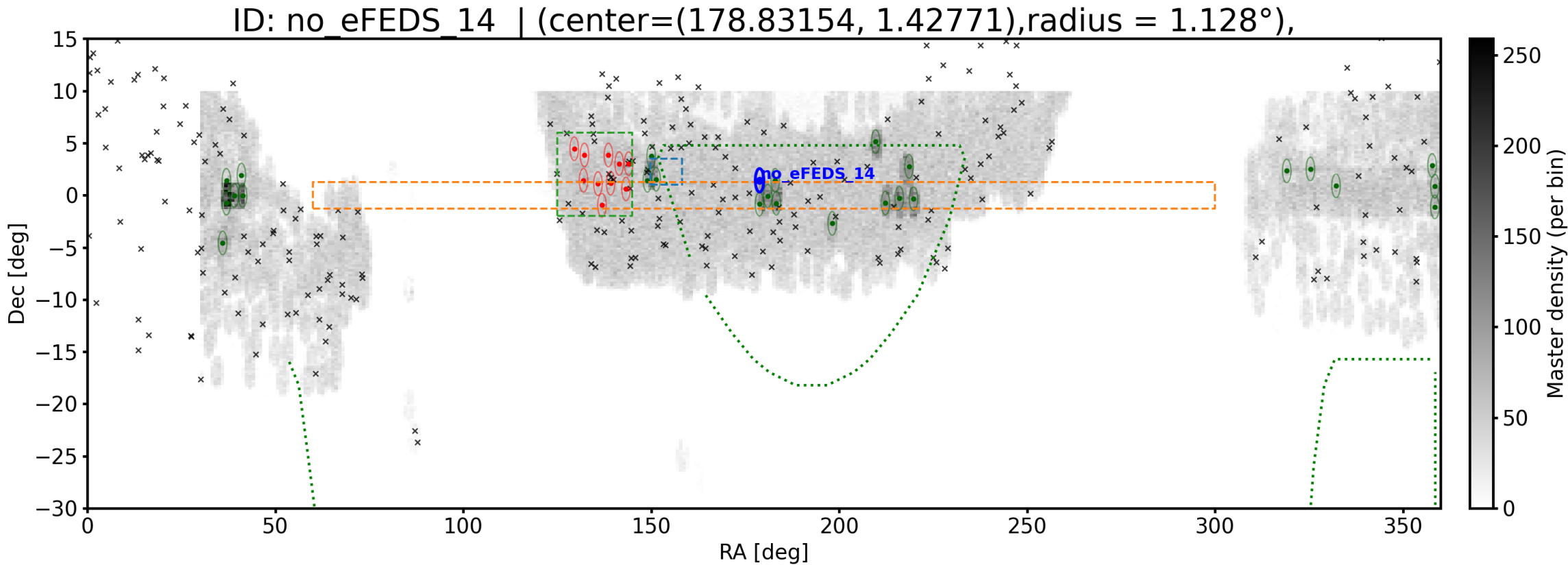


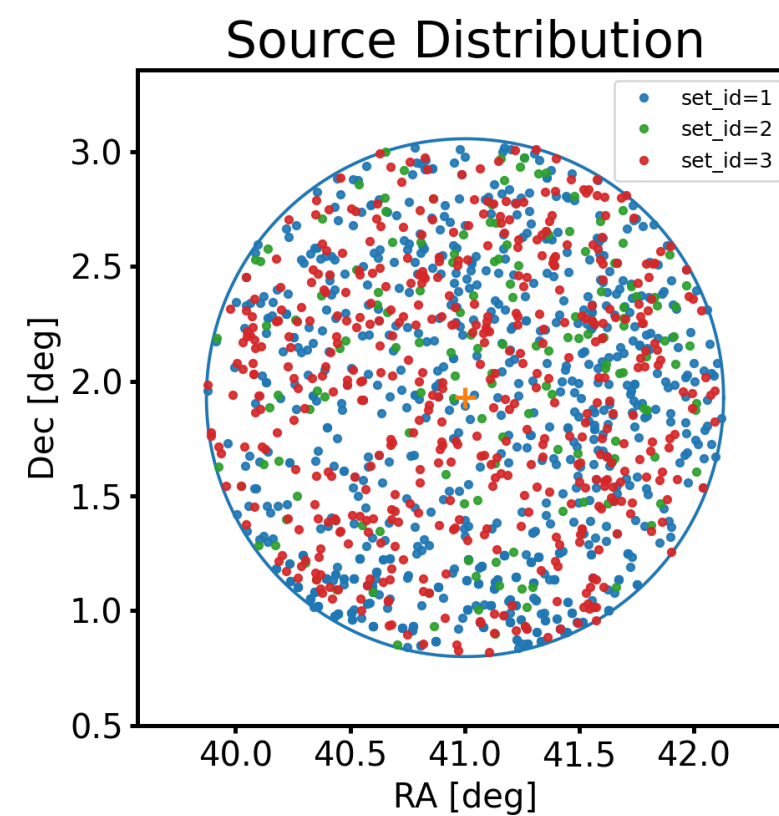
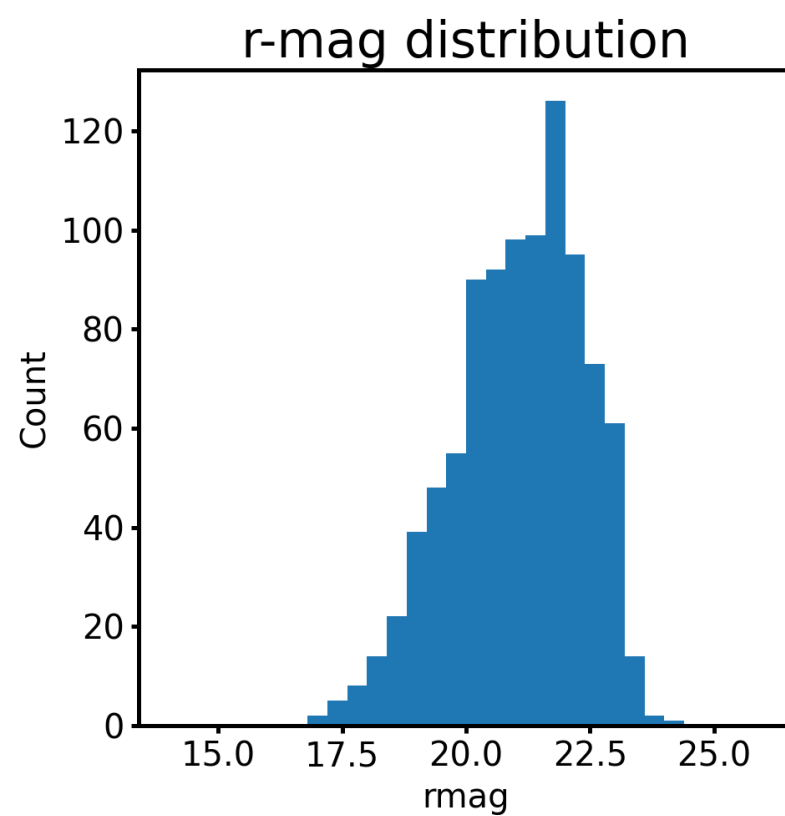
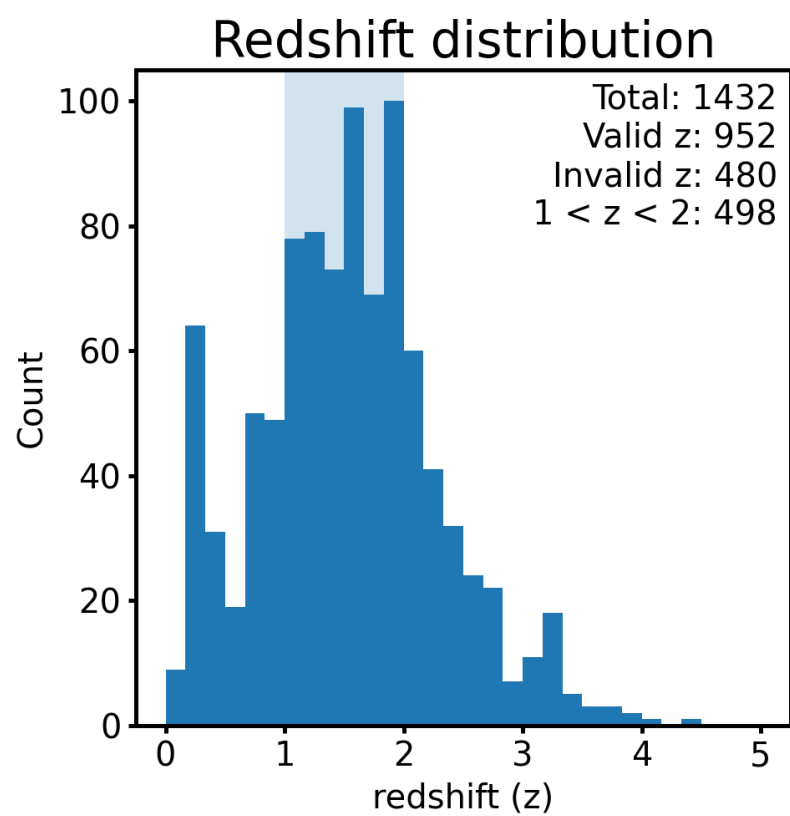
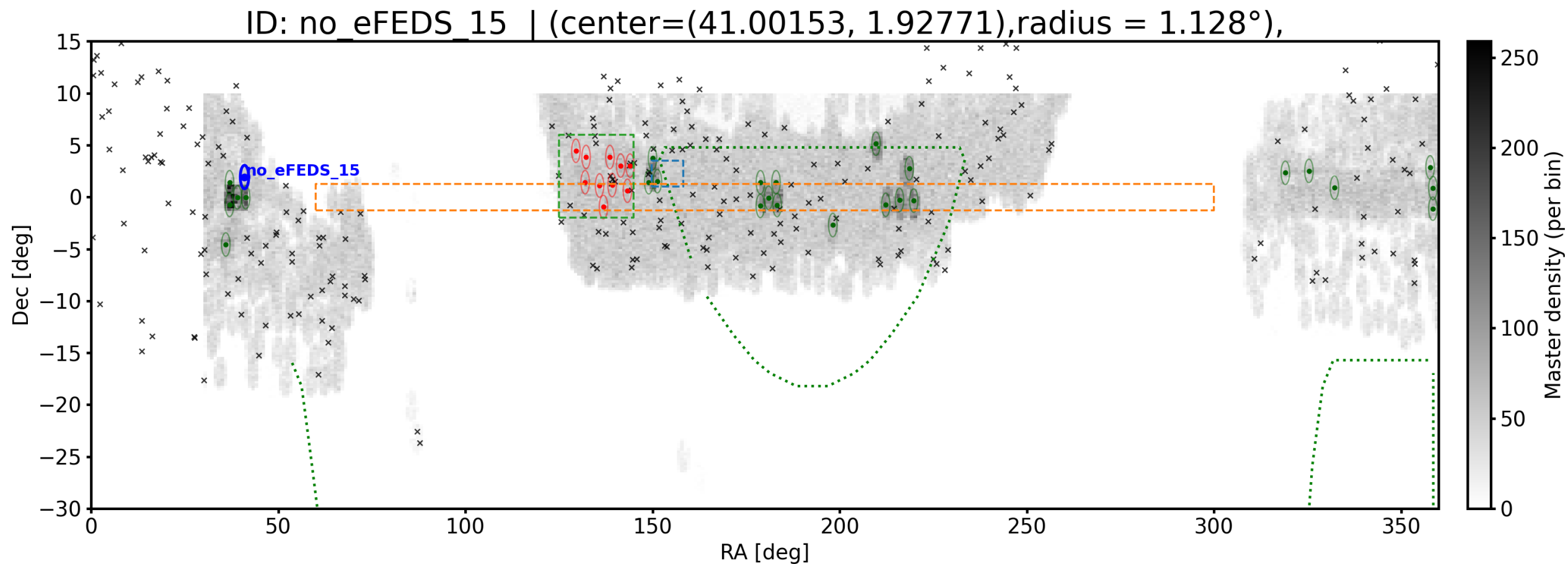












Old Selection

Summary Table : Inside eFEDS

ID	RA_center	DEC_center	sel1 total	sel1 common	sel1 noncommon	sel2 total	sel3 total	sel3 eRASS	sel3 unwise	grand total
eFEDS_1	144.001535	3.027709	800	189	611	220	480	136	344	1500
eFEDS_2	141.501535	3.027709	800	181	619	287	480	134	346	1567
eFEDS_3	132.001535	1.427709	800	311	489	282	480	137	343	1562
eFEDS_4	139.201535	1.155000	800	390	410	310	480	133	347	1590
eFEDS_5	136.901535	-0.957229	800	137	663	320	480	125	355	1600
eFEDS_6	143.301535	0.609000	800	165	635	295	480	114	366	1575
eFEDS_7	129.530000	4.454000	800	102	698	299	480	133	347	1579
eFEDS_8	132.250000	3.859000	800	105	695	282	480	129	351	1562
eFEDS_9	135.850000	1.111000	800	185	615	291	480	122	358	1571
eFEDS_10	138.630000	3.852000	800	138	662	273	480	131	349	1553

Field	RA_center	DEC_center	sel1 total	sel1 common	sel1 common	sel2 total	sel3 total	sel3 ePASS	sel3 unwise	grand total	
no_eFEDS_1	37.001535	-0.757229	800	0	800	230	480	0	480	1510	
no_eFEDS_2	41.301535	-0.027229	800	0	800	202	480	0	480	1482	
no_eFEDS_3	39.101535	-0.027229	800	0	800	234	480	0	480	1514	
no_eFEDS_4	183.301535	-0.772291	800	0	800	320	480	198	282	1600	PAQS/S82
no_eFEDS_5	181.001535	-0.072291	800	0	800	320	480	165	315	1600	PAQS/S82
no_eFEDS_6	37.001535	1.427709	800	0	800	236	480	0	480	1516	
no_eFEDS_7	209.701535	5.137709	800	0	800	320	480	186	294	1600	
no_eFEDS_8	150.001535	3.727709	800	0	800	292	480	106	374	1572	COSMOS
no_eFEDS_9	178.801535	-0.857229	800	0	800	320	480	156	324	1600	PAQS
no_eFEDS_10	151.301535	1.527709	800	0	800	320	480	130	350	1600	COSMOS
no_eFEDS_11	149.001535	1.427709	800	0	800	320	480	107	373	1600	COSMOS
no_eFEDS_12	36.001535	-4.572291	800	0	800	320	480	0	480	1600	
no_eFEDS_13	183.001535	1.427709	800	0	800	320	480	182	298	1600	PAQS
no_eFEDS_14	178.831535	1.427709	800	0	800	320	480	161	319	1600	
no_eFEDS_15	41.001535	1.927709	800	0	800	152	480	0	480	1432	
no_eFEDS_16	198.170000	-2.704170	800	0	800	288	480	186	294	1568	PAQS
no_eFEDS_17	218.603243	2.758373	800	0	800	320	480	206	274	1600	PAQS
no_eFEDS_18	219.790775	-0.341460	800	0	800	320	480	238	242	1600	PAQS/S82
no_eFEDS_19	212.314362	-0.749672	800	0	800	320	480	184	296	1600	PAQS/S82
no_eFEDS_20	215.985987	-0.277543	800	0	800	320	480	238	242	1600	PAQS/S82
no_eFEDS_21	332.180000	0.900000	800	0	800	226	480	0	480	1506	
no_eFEDS_22	319.060000	2.332000	800	0	800	259	480	0	480	1539	
no_eFEDS_23	325.360000	2.513000	800	0	800	256	480	0	480	1536	
no_eFEDS_24	358.480000	0.873000	800	0	800	194	480	0	480	1474	
no_eFEDS_25	358.480000	-1.122000	800	0	800	181	480	0	480	1461	
no_eFEDS_26	357.740000	2.861000	800	0	800	103	480	0	480	1383	

Field	RA_center	DEC_center	set1_#	set1_%	set2_#	set2_%	set3_#	set3_%	total_sel	target
no_eFEDS_1	37.001535	-0.757229	800	53.0	230	15.2	480	31.8	1510	1600
no_eFEDS_2	41.301535	-0.027229	800	54.0	202	13.6	480	32.4	1482	1600
no_eFEDS_3	39.101535	-0.027229	800	52.8	234	15.5	480	31.7	1514	1600
no_eFEDS_4	183.301535	-0.772291	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_5	181.001535	-0.072291	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_6	37.001535	1.427709	800	52.8	236	15.6	480	31.7	1516	1600
no_eFEDS_7	209.701535	5.137709	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_8	150.001535	3.727709	800	50.9	292	18.6	480	30.5	1572	1600
no_eFEDS_9	178.801535	-0.857229	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_10	151.301535	1.527709	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_11	149.001535	1.427709	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_12	36.001535	-4.572291	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_13	183.001535	1.427709	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_14	178.831535	1.427709	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_15	41.001535	1.927709	800	55.9	152	10.6	480	33.5	1432	1600
no_eFEDS_16	198.170000	-2.704170	800	51.0	288	18.4	480	30.6	1568	1600
no_eFEDS_17	218.603243	2.758373	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_18	219.790775	-0.341460	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_19	212.314362	-0.749672	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_20	215.985987	-0.277543	800	50.0	320	20.0	480	30.0	1600	1600
no_eFEDS_21	332.180000	0.900000	800	53.1	226	15.0	480	31.9	1506	1600
no_eFEDS_22	319.060000	2.332000	800	52.0	259	16.8	480	31.2	1539	1600
no_eFEDS_23	325.360000	2.513000	800	52.1	256	16.7	480	31.3	1536	1600
no_eFEDS_24	358.480000	0.873000	800	54.3	194	13.2	480	32.6	1474	1600
no_eFEDS_25	358.480000	-1.122000	800	54.8	181	12.4	480	32.9	1461	1600
no_eFEDS_26	357.740000	2.861000	800	57.8	103	7.4	480	34.7	1383	1600